

Non-linear Conjugate Gradient With Restart Also Escapes Saddle Points

Working document — full proofs and rationale

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Abstract

We study a perturbed, restarted variant of the non-linear conjugate gradient (NCG) method and show that, like perturbed gradient descent, it escapes strict saddle points and converges to an approximate second-order stationary point. The analysis combines per-step decrease guarantees, an improve-or-localize argument valid for both restarted and non-restarted iterations, a linearization of the dynamics around the snapshot point, an anti-concentration estimate for the random perturbation, an induction maintaining three coupled invariants over the escape window, and a contradiction argument converting geometric growth into guaranteed function decrease. A final section resolves the resulting constraint system exactly and derives the complexity bound, together with its current limitations.

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1 Introduction

We follow the framework of [2] and [3] and adapt it to the non-linear conjugate gradient method with restart, in order to show that this method also escapes strict saddle points and produces an approximate second-order stationary point. The starting point for the descent estimates is the NCG complexity analysis of [1]; the saddle-avoidance mechanism is borrowed, at a high level, from the perturbed gradient descent analyses of [2, 3] and from the dynamical-systems viewpoint of [4]. Our setup is comparable to the one of [3], since NCG is a momentum method, just like accelerated gradient descent (AGD). However, we depart from their proof technique by studying a *single trajectory*, whereas they study two coupled trajectories of the same process and conclude from the growth of their difference.

The document is organized around the proof story of Section 4: nine steps, each with a one-paragraph rationale (*what* it establishes, *why* the argument needs it, and *where* it is consumed), followed by the corresponding section with full proofs.

2 Preliminaries

In this section, we introduce our notation, and present the definitions, assumptions, and elementary consequences used throughout this document.

2.1 Notation

For vectors, we use $\|\cdot\|$ to denote the ℓ_2 -norm. We use $\lambda_{\max}(\cdot)$ and $\lambda_{\min}(\cdot)$ to denote the largest and smallest eigenvalues respectively. For a function $f: \mathbb{R}^d \rightarrow \mathbb{R}$, we use $\nabla f(\cdot)$ and $\nabla^2 f(\cdot)$ to denote its gradient and Hessian, and f^* to denote the global minimum of f . We let $\mathcal{B}^d(x, r)$ denote the d -dimensional ball centered at x with radius r , and $\mathcal{U}(S)$ the uniform distribution over a set S .

2.2 Setting

We consider the unconstrained minimization problem

$$\min_{x \in \mathbb{R}^d} f(x), \tag{1}$$

where $f: \mathbb{R}^d \rightarrow \mathbb{R}$ is nonconvex and whose saddle points are strict.

Definition 1 (Strict saddle point). A strict saddle point is a saddle point x^* that has at least one direction of negative curvature, that is,

$$\lambda_{\min}(\nabla^2 f(x^*)) < 0. \quad (2)$$

The algorithm certifies approximate second-order stationarity in the following sense.

Definition 2 (Approximate second-order stationary point). Given tolerances $\varepsilon_g, \varepsilon_H > 0$, a point $x \in \mathbb{R}^d$ is an $(\varepsilon_g, \varepsilon_H)$ -second-order stationary point (SOSP) if

$$\|\nabla f(x)\| \leq \varepsilon_g \quad \text{and} \quad \lambda_{\min}(\nabla^2 f(x)) \geq -\varepsilon_H. \quad (3)$$

We make the following standing assumptions.

Assumption 1 (Gradient Lipschitz). The function f is continuously differentiable on $\mathbb{R}^d$, and its gradient ∇f is L_1 -Lipschitz continuous, with $L_1 > 0$.

Assumption 2 (Hessian Lipschitz). The function f is twice continuously differentiable on $\mathbb{R}^d$, and its Hessian $\nabla^2 f$ is L_2 -Lipschitz continuous, with $L_2 > 0$.

2.3 Elementary consequences of the assumptions

Both consequences below are used repeatedly; we record them once with full proofs.

Lemma 1 (Descent lemma). *Under Assumption 1, for all $x, y \in \mathbb{R}^d$,*

$$f(y) \leq f(x) + \nabla f(x)^\top (y - x) + \frac{L_1}{2} \|y - x\|^2. \quad (4)$$

Proof. Let $\varphi(t) := f(x + t(y - x))$ for $t \in [0, 1]$, so that $\varphi'(t) = \nabla f(x + t(y - x))^\top (y - x)$. By the fundamental theorem of calculus,

$$f(y) - f(x) = \int_0^1 \varphi'(t) dt = \nabla f(x)^\top (y - x) + \int_0^1 (\nabla f(x + t(y - x)) - \nabla f(x))^\top (y - x) dt.$$

By Cauchy–Schwarz and Assumption 1, the integrand is bounded by $L_1 t \|y - x\|^2$, and $\int_0^1 L_1 t dt = L_1/2$, giving (4). $\square$

Lemma 2 (Spectral bound on the Hessian). *Under Assumptions 1 and 2, for every $x \in \mathbb{R}^d$, $\|\nabla^2 f(x)\| \leq L_1$; in particular $\lambda_{\min}(\nabla^2 f(x)) \geq -L_1$ everywhere.*

Proof. Fix x and a unit vector v . By Assumption 1, $\|\nabla f(x + tv) - \nabla f(x)\| \leq L_1 |t|$ for all t , so

$$\|\nabla^2 f(x) v\| = \lim_{t \rightarrow 0} \frac{\|\nabla f(x + tv) - \nabla f(x)\|}{|t|} \leq L_1.$$

Taking the supremum over unit v gives $\|\nabla^2 f(x)\| \leq L_1$; since $\nabla^2 f(x)$ is symmetric, all its eigenvalues lie in $[-L_1, L_1]$. $\square$

Remark 1 (Meaningful range of ε_H). By Lemma 2, the curvature condition $\lambda_{\min}(\nabla^2 f(x)) \leq -\varepsilon_H$ can hold at *some* point only if $\varepsilon_H \leq L_1$. For $\varepsilon_H > L_1$ the perturbation trigger of Algorithm 1 can never fire and the SOSP certificate (3) holds automatically at every ε_g -stationary point. This elementary observation becomes important when solving the constraints system.

3 The Perturbed Restarted NCG Algorithm

We present a perturbed variant of the NCG method with restart, analogous to the perturbed gradient descent of [2]. The algorithm augments the usual NCG iteration with four ingredients absent from the plain method: a perturbation radius r , an escape window T_{perturb} , a function-value progress threshold F_{thresh} , and a target failure probability δ . The idea is straightforward: when the norm of the current gradient is small ($\|g_k\| \leq \varepsilon_g$), the current point is a candidate SOSP, and a random perturbation is injected to test whether it is a SOSP or a strict saddle point.

Algorithm 1 Perturbed Restarted Non-linear Conjugate Gradient (PR-NCG)

Require: $x_0 \in \mathbb{R}^d$; NCG parameters $\sigma \in (0, 1]$, $\kappa \geq 1$, $p \geq 0$, $q \geq 0$; tolerances $\varepsilon_g > 0$, $\varepsilon_H > 0$; perturbation radius $r > 0$, escape window $T_{\text{perturb}} \in \mathbb{N}$, progress threshold $F_{\text{thresh}} > 0$, failure probability $\delta \in (0, 1)$.

- 1: $g_0 \leftarrow \nabla f(x_0)$, $d_0 \leftarrow -g_0$
- 2: $t_{\text{noise}} \leftarrow -T_{\text{perturb}} - 1$ ▷ allows an immediate perturbation
- 3: $\tilde{x} \leftarrow x_0$, $\hat{f} \leftarrow f(x_0)$ ▷ last snapshot
- 4: **for** $k = 0, 1, 2, \dots$ **do**
- 5: $g_k \leftarrow \nabla f(x_k)$
- 6: **if** $\|g_k\| \leq \varepsilon_g$ **and** $k - t_{\text{noise}} > T_{\text{perturb}}$ **then** ▷ perturbation trigger (replaces NCD)
- 7: $\tilde{x} \leftarrow x_k$, $\hat{f} \leftarrow f(x_k)$ ▷ save snapshot
- 8: sample $\xi \sim \mathcal{U}(\mathcal{B}^d(0, r))$
- 9: $x_k \leftarrow x_k + \xi$
- 10: $g_k \leftarrow \nabla f(x_k)$ ▷ recompute after perturbation
- 11: $d_k \leftarrow -g_k$ ▷ reset momentum
- 12: $t_{\text{noise}} \leftarrow k$
- 13: **end if**
- 14: **if** $k - t_{\text{noise}} = T_{\text{perturb}}$ **and** $f(x_k) - \hat{f} > -F_{\text{thresh}}$ **then** ▷ termination check
- 15: **return** $\tilde{x}$ as an approximate $(\varepsilon_g, \varepsilon_H)$ -2OSP
- 16: **end if**
- 17: /* NCG step */ ▷ restarted direction
- 18: **if** $d_k = -g_k$ **then**
- 19: $\alpha_k \leftarrow \alpha_{\mathcal{R}}^* = 1/L_1$
- 20: **else**
- 21: $\alpha_k \leftarrow \alpha_{\mathcal{N}}^* = \frac{\sigma}{L_1 \kappa^2} \|g_k\|^{1+p-2q}$
- 22: **end if**
- 23: $x_{k+1} \leftarrow x_k + \alpha_k d_k$
- 24: $g_{k+1} \leftarrow \nabla f(x_{k+1})$
- 25: choose β_{k+1}
- 26: $d_{k+1} \leftarrow -g_{k+1} + \beta_{k+1} d_k$
- 27: **if** $g_{k+1}^\top d_{k+1} \geq -\sigma \|g_{k+1}\|^{1+p}$ **or** $\|d_{k+1}\| \geq \kappa \|g_{k+1}\|^q$ **then** ▷ restart condition
- 28: $d_{k+1} \leftarrow -g_{k+1}$
- 29: **end if**
- 30: **end for**

We write $\mathcal{N}$ for the set of non-restarted iterations (the kept NCG direction satisfies $g_k^\top d_k < -\sigma \|g_k\|^{1+p}$ and $\|d_k\| < \kappa \|g_k\|^q$, by the complement of the restart test) and $\mathcal{R}$ for the restarted ones ($d_k = -g_k$).

Assumption 3 (Exponent matching). Throughout the escape analysis we take $1 + p = 2q$ with

$q = 1$, i.e. $p = q = 1$. Under this assumption the non-restarted stepsize is the *constant*

$$\alpha_{\mathcal{N}}^* = \frac{\sigma}{L_1 \kappa^2}, \quad \text{with} \quad \alpha_{\mathcal{N}}^* L_1 = \frac{\sigma}{\kappa^2} \leq 1,$$

since $\sigma \leq 1 \leq \kappa^2$. We write

$$\alpha_{\min} := \frac{\sigma}{L_1 \kappa^2}, \quad \text{so that every stepsize satisfies} \quad \alpha_k \in \left\{ \alpha_{\min}, \frac{1}{L_1} \right\} \subset \left[\alpha_{\min}, \frac{1}{L_1} \right], \quad \alpha_k L_1 \leq 1. \quad (5)$$

The restart test then reads $\|d_{k+1}\| \geq \kappa \|g_{k+1}\|$ and $g_{k+1}^\top d_{k+1} \geq -\sigma \|g_{k+1}\|^2$.

Remark 2 (Changelog: removal of the in-window stepsize safeguard). An earlier version of Algorithm 1 had the stepsize at $\alpha = 1/L_1$ inside the escape window, for *all* directions. This created a third iteration type: NCG direction with $\alpha = 1/L_1$, which was covered by neither case of Lemma 5, and for which the per-step decrease can fail: the descent lemma only gives $f(x_k) - f(x_{k+1}) \geq \frac{\sigma}{L_1} \|g_k\|^{1+p} - \frac{\kappa^2}{2L_1} \|g_k\|^{2q}$, whose sign is uncontrolled in general. Since Lemma 10 applies Lemma 5 to in-window steps, this was a gap.

Resolution adopted. Under Assumption 3 the stepsize $\alpha_{\mathcal{N}}^*$ is constant, so the safeguard is unnecessary: the window runs the algorithm's ordinary stepsize rule, every in-window step is one of the two cases of Lemma 5 (restarted with $1/L_1$, or non-restarted with $\alpha_{\mathcal{N}}^*$), and the uniform constant $2/L_1$ applies verbatim. The escape recurrence becomes variable-stepsize with $\alpha_k \in [\alpha_{\min}, 1/L_1]$; the growth rate is controlled by $\alpha_{\min}$, costing the absolute factor κ^2/σ in T and κ^4/σ^2 in N_{iter} , with no change to the $(\varepsilon_H, d, \delta)$ dependence.

Alternatives not adopted. (a) Keeping $\alpha = 1/L_1$ in-window and requiring $\kappa^2 < 2\sigma$ also restores a per-step bound, with the non-uniform constant $2c_w/L_1$, $c_w = \kappa^2/(2\sigma - \kappa^2)$; valid (localization only needs the maximum of the per-step constants), but it confines (σ, κ) to $\kappa \in [1, \sqrt{2\sigma})$. (b) For general (p, q) , the safeguard $\alpha_k = \min\{1/L_1, \alpha_{\mathcal{N}}^*\}$ restores the bound but makes the in-window growth rate depend on $\underline{g}^{1+p-2q}$, degrading the final rate; not pursued.

3.1 Perturbation as a saddle detector

The perturbation plays the role of a strict-saddle detector. The intuition is that when the gradient is small, we are at a candidate SOSP, and we perturb to get one of the following two conclusions:

- If we perturb a point close to a strict saddle and let the algorithm run, the iterate moves a lot, because the negative-curvature direction amplifies the perturbation exponentially, and the function value drops significantly.
- If we perturb a point close to a local minimizer (a genuine SOSP) and let the algorithm run, the iterate stays roughly the same and the function value stagnates.

Hence the test “did f drop substantially after the perturbation?” answers the question “was I at a strict saddle?”. This permits us to certify convergence to an SOSP without any explicit computation of the Hessian matrix: condition $\|g_k\| \leq \varepsilon_g$ is trivial to check, while the curvature condition $\lambda_{\min}(\nabla^2 f(x_k)) \geq -\varepsilon_H$ is certified *without* an explicit Hessian computation, through the perturbation test described above.

4 Proof Story (How to Read This Document)

The argument decomposes into nine steps. Steps I–II are generic descent facts about the algorithm; Steps III–VI build the toolbox for the near-saddle analysis; Step VII is the heart of the proof (the induction); Step VIII converts it into the escape theorem; Step IX resolves the constants and counts iterations. For each step we record *what* it establishes, *why* the argument needs it, and *where* it is proved.

Step I — Decrease lemmas (Section 5). Basic: each regime (restarted, non-restarted) gives a per-step decrease guarantee, with the stepsize α chosen to make the decrease optimal in that regime. *Why:* these are the raw material of every function-value accounting in the document, in particular of Step II.

Step II — Improve or localize (Section 6). We prove

$$\sum_{k=0}^{T-1} \|x_{k+1} - x_k\|^2 \leq \frac{2}{L_1} (f(x_0) - f(x_T)).$$

There are two readings of this inequality. (1) *Improve:* if the iterates have moved a lot (left-hand side large), the function value must have dropped a lot (right-hand side large). (2) *Localize:* if the function value has not dropped much (small right-hand side), the iterates cannot have moved much (small left-hand side). This is a dichotomy: at any moment, either we have improved by a meaningful amount, or we are stuck near where we started. *Why:* the entire escape proof works inside the *localize* branch and derives a contradiction there; the dichotomy then forces the *improve* branch.

Step III — Linearization around $\tilde{x}$ (Section 7.1). Here we linearize in order to perform a spectral analysis (i.e. get a linear system and compute eigenvalues). The whole idea is to show that the system has an eigenvalue $\mu_+ > 1$: since the element of study is $u_k = x_k - \tilde{x}$, if $\mu_+ > 1$, the iterate escapes the stationary point along the negative-curvature direction. We let $c_k := \langle u_k, v_{\min} \rangle$ and study its recurrence to get geometric growth. The recurrence is

$$|c_{k+1}| \geq (1 + \alpha \varepsilon_H) |c_k| - \alpha \beta_k |\eta_k| - \alpha \varepsilon_g - \alpha \|\tilde{e}_k\|,$$

with $\eta_k = \langle d_{k-1}, v_{\min} \rangle$ a momentum term and $\tilde{e}_k = \nabla f(x_k) - g_{\tilde{x}} - H(x_k - \tilde{x})$ an error term. The rest of the proof exists to control the other terms in the recurrence (momentum and quadratic error) and to show that escaping along negative curvature has high probability.

Step IV — Quadratic model error (Section 7.2). Straightforward: bound $\|\tilde{e}_k\|$ above, which corresponds to the error between the real gradient and the quadratically approximated one. *Why:* the recurrence of Step III is only useful if this error cannot cancel the geometric push.

Step V — Probabilistic anti-concentration (Section 7.4). Also a straightforward step. The idea is to show that the perturbation has a non-negligible component along the negative-curvature direction, with high probability. *Why:* the geometric growth $|c_k| \gtrsim \mu_+^k |c_0|$ is vacuous if $|c_0|$ is zero or tiny; this step supplies the initial signal $|c_0| \geq s_\delta$.

Step VI — Lower bound on $\underline{g}$ (Fletcher–Reeves specific) (Section 7.5). This step exists to control the momentum term. Since $\beta_k^{\text{FR}} = \|g_k\|^2 / \|g_{k-1}\|^2$, we have to bound $\beta^{\text{FR}} \leq \bar{g}^2 / \underline{g}^2$, which requires a *lower* bound on gradient norms. Any other β formula would mean redoing this step; heavy ball or Nesterov do not need it, because their momentum coefficient is constant. *Why:* this is the FR-specific price of the analysis, and the source of the final regime restriction.

Step VII — Induction for geometric growth (Section 7.7). We want $|c_k| \geq (1 + \gamma \alpha \varepsilon_H)^k |c_0|$. The issue is that the obstacle terms in the recurrence depend on the past: $\|\tilde{e}_k\| \leq \frac{L_2}{2} \|u_k\|^2$ depends on $u_k = x_k - \tilde{x}$; β_k depends on $\|g_{k-1}\|$; $|\eta_k| = |\langle d_{k-1}, v_{\min} \rangle|$ depends on previous directions. The solution is an induction, which lets us maintain a set of properties

simultaneously up to time k and prove them at time $k + 1$. From the recurrence, the three things we need are: (1) how far the iterate is ($\|x_k - \tilde{x}\| \leq \dots$); (2) how big the gradient is ($\|g_k\| \geq g$); (3) how big $|c_k|$ is ($|c_k| \geq \dots |c_0|$). The whole idea is then driven by improve-or-localize: (a) the function value drops by at least F somewhere in $\{0, \dots, T\}$: this is what we try to prove; (b) the function value never drops by that much, in which case Step II tells us the iterates stay close to x_0 ($\|x_k - x_0\| \leq S$, $S = \sqrt{2FT/L_1}$). We work in case (b) and derive a contradiction, so we must be in case (a). Case (b) is the bad case and we formalize it by the localization event $\mathcal{L}_T = \{\max_{0 \leq k \leq T} (f(x_0) - f(x_k)) < F\}$.

Step VIII — Escape theorem (Section 7.8). Using invariant (3) of the induction, show $|c_T| \geq (1 + \gamma\alpha\varepsilon_H)^T s_\delta$, and using invariant (1), $|c_T| \leq \|u_T\| \leq S + R^*$. For a suitable T , $(1 + \gamma\alpha\varepsilon_H)^T s_\delta > S + R^*$: we have the contradiction $|c_T| \leq S + R^*$ and $|c_T| > S + R^*$. Hence $\mathcal{G}_0 \cap \mathcal{L}_T = \emptyset$ (since we have been conditioning on $\mathcal{G}_0$). This means we are in the first case of improve-or-localize, and the function value drops by at least F within the escape window with high probability.

5 Step I — Decrease Lemmas

Rationale. Each iteration type comes with a stepsize chosen to make the guaranteed decrease optimal for that type. The two lemmas below are the per-step guarantees; they feed the improve-or-localize inequality of Step II, and (through it) the entire function-value accounting of the escape proof.

Lemma 3 (Decrease on non-restarted iterations). *For every non-restarted iteration $k \in \mathcal{N}$ (so that $g_k^\top d_k \leq -\sigma \|g_k\|^{1+p}$, $\|d_k\| \leq \kappa \|g_k\|^q$, and $\alpha_k = \alpha_N^* = \frac{\sigma}{L_1 \kappa^2} \|g_k\|^{1+p-2q}$),*

$$f(x_k) - f(x_{k+1}) \geq c_N \|g_k\|^{2(1+p-q)}, \quad c_N = \frac{\sigma^2}{2L_1 \kappa^2}. \quad (6)$$

Proof. This is the fixed-stepsize NCG decrease guarantee of [1]; we reproduce the computation for completeness. Apply the descent lemma (Lemma 1) at $x = x_k$, $y = x_{k+1} = x_k + \alpha_k d_k$:

$$f(x_{k+1}) \leq f(x_k) + \alpha_k g_k^\top d_k + \frac{L_1}{2} \alpha_k^2 \|d_k\|^2.$$

By the two properties of a kept direction, $g_k^\top d_k \leq -\sigma \|g_k\|^{1+p}$ and $\|d_k\|^2 \leq \kappa^2 \|g_k\|^{2q}$, so

$$f(x_k) - f(x_{k+1}) \geq \alpha_k \sigma \|g_k\|^{1+p} - \frac{L_1}{2} \alpha_k^2 \kappa^2 \|g_k\|^{2q}.$$

Substituting $\alpha_k = \frac{\sigma}{L_1 \kappa^2} \|g_k\|^{1+p-2q}$: the first term equals $\frac{\sigma^2}{L_1 \kappa^2} \|g_k\|^{(1+p-2q)+(1+p)} = \frac{\sigma^2}{L_1 \kappa^2} \|g_k\|^{2(1+p-q)}$, and the second term equals $\frac{L_1}{2} \cdot \frac{\sigma^2}{L_1^2 \kappa^4} \|g_k\|^{2(1+p-2q)} \cdot \kappa^2 \|g_k\|^{2q} = \frac{\sigma^2}{2L_1 \kappa^2} \|g_k\|^{2(1+p-q)}$. Subtracting,

$$f(x_k) - f(x_{k+1}) \geq \frac{\sigma^2}{L_1 \kappa^2} \left(1 - \frac{1}{2}\right) \|g_k\|^{2(1+p-q)} = \frac{\sigma^2}{2L_1 \kappa^2} \|g_k\|^{2(1+p-q)}. \quad \square$$

Remark 3 (Optimality of α_N^*). The right-hand side of the intermediate bound, $\alpha \sigma \|g\|^{1+p} - \frac{L_1 \kappa^2}{2} \alpha^2 \|g\|^{2q}$, is a downward parabola in α , maximized exactly at $\alpha = \frac{\sigma}{L_1 \kappa^2} \|g\|^{1+p-2q} = \alpha_N^*$.

Lemma 4 (Decrease on restarted iterations). *For every restarted iteration $k \in \mathcal{R}$ (so that $d_k = -g_k$ and $\alpha_k = 1/L_1$),*

$$f(x_k) - f(x_{k+1}) \geq c_R \|g_k\|^2, \quad c_R = \frac{1}{2L_1}. \quad (7)$$

Proof. Apply Lemma 1 with $y - x = -\frac{1}{L_1}g_k$:

$$f(x_{k+1}) \leq f(x_k) - \frac{1}{L_1} \|g_k\|^2 + \frac{L_1}{2} \cdot \frac{1}{L_1^2} \|g_k\|^2 = f(x_k) - \frac{1}{2L_1} \|g_k\|^2.$$

Rearranging gives (7). As in the previous remark, $\alpha = 1/L_1$ maximizes $\alpha \|g\|^2 - \frac{L_1}{2}\alpha^2 \|g\|^2$, so the restarted stepsize is also decrease-optimal for its regime. $\square$

6 Step II — Improve or Localize

Rationale. The goal of this section is to bound the total squared displacement $\sum_k \|x_{k+1} - x_k\|^2$ by the total function decrease, uniformly over restarted and non-restarted iterations. We treat the two cases separately and find the *same* per-step bound, which we then telescope. The resulting inequality admits the two readings of Step II in Section 4 (improve / localize); the localize reading is the one consumed by the escape proof.

6.1 Non-restarted iterations ($k \in \mathcal{N}$)

For a non-restarted iteration the restart condition guarantees $\|d_k\| \leq \kappa \|g_k\|^q$, so

$$\begin{aligned} \|x_{k+1} - x_k\|^2 &= \alpha_k^2 \|d_k\|^2 \leq \alpha_k^2 \kappa^2 \|g_k\|^{2q} \\ &= \frac{\sigma^2}{L_1^2 \kappa^4} \|g_k\|^{2(1+p-2q)} \kappa^2 \|g_k\|^{2q} = \left(\frac{\sigma}{L_1 \kappa} \right)^2 \|g_k\|^{2(1+p-q)}, \end{aligned} \quad (8)$$

where we used $\alpha_k = \frac{\sigma}{L_1 \kappa^2} \|g_k\|^{1+p-2q}$ and combined the exponents $2(1+p-2q) + 2q = 2(1+p-q)$.

From Lemma 3, $c_{\mathcal{N}} \|g_k\|^{2(1+p-q)} \leq f(x_k) - f(x_{k+1})$, that is,

$$\|g_k\|^{2(1+p-q)} \leq \frac{2L_1 \kappa^2}{\sigma^2} (f(x_k) - f(x_{k+1})). \quad (9)$$

Substituting (9) into (8) yields

$$\|x_{k+1} - x_k\|^2 \leq \frac{\sigma^2}{L_1^2 \kappa^2} \cdot \frac{2L_1 \kappa^2}{\sigma^2} (f(x_k) - f(x_{k+1})) = \frac{2}{L_1} (f(x_k) - f(x_{k+1})). \quad (10)$$

6.2 Restarted iterations ($k \in \mathcal{R}$)

For a restarted iteration $d_k = -g_k$ and $\alpha_k = 1/L_1$, hence

$$\|x_{k+1} - x_k\|^2 = \alpha_k^2 \|d_k\|^2 = \alpha_k^2 \|g_k\|^2 = \frac{1}{L_1^2} \|g_k\|^2. \quad (11)$$

From Lemma 4, $\|g_k\|^2 \leq \frac{1}{c_{\mathcal{R}}} (f(x_k) - f(x_{k+1})) = 2L_1 (f(x_k) - f(x_{k+1}))$, so

$$\|x_{k+1} - x_k\|^2 \leq \frac{1}{L_1^2} \cdot 2L_1 (f(x_k) - f(x_{k+1})) = \frac{2}{L_1} (f(x_k) - f(x_{k+1})). \quad (12)$$

6.3 The uniform improve-or-localize lemma

Lemma 5 (Improve or localize). *For every iteration k taken with the optimal stepsize α^* ,*

$$\|x_{k+1} - x_k\|^2 \leq \frac{2}{L_1} (f(x_k) - f(x_{k+1})). \quad (13)$$

In particular each such step decreases f . Consequently, telescoping over any window of consecutive such steps $k = 0, \dots, T-1$,

$$\sum_{k=0}^{T-1} \|x_{k+1} - x_k\|^2 \leq \frac{2}{L_1} (f(x_0) - f(x_T)) \leq \frac{2}{L_1} (f(x_0) - f^*). \quad (14)$$

Proof. The per-step bound (13) is (10) for $k \in \mathcal{N}$ and (12) for $k \in \mathcal{R}$; the two regimes give the same constant. The left-hand side of (13) is nonnegative, so each per-step decrease $f(x_k) - f(x_{k+1})$ is nonnegative. Summing (13) over $k = 0, \dots, T-1$ and telescoping the right-hand side gives (14). $\square$

Remark 4 (Interpretation: a dichotomy). Lemma 5 is a dichotomy. At any moment, either the function value has decreased by a meaningful amount (*improve*), or the iterates did not move much and we are stuck near where we started (*localize*). The aim of Section 7 is to upgrade this dichotomy near a strict saddle into the statement: *at a strict saddle, we always improve*. If we can prove that, then strict saddles are not an obstruction, and a failure to improve at some iteration can only mean we are at a (local) minimizer.

Remark 5 (Flagged gap: the in-window stepsize is a third case). Lemma 5 covers exactly two iteration types: non-restarted steps with $\alpha = \alpha_{\mathcal{N}}^*$, and restarted steps with $\alpha = 1/L_1$. Inside the escape window, however, Algorithm 1 takes $\alpha_k = 1/L_1$ with *NCG directions* — a third combination, covered by neither case. For this combination the per-step decrease can genuinely fail: the descent lemma gives

$$f(x_k) - f(x_{k+1}) \geq \frac{\sigma}{L_1} \|g_k\|^{1+p} - \frac{\kappa^2}{2L_1} \|g_k\|^{2q},$$

whose sign depends on $\|g_k\|$. Since Lemma 10 below applies Lemma 5 to in-window steps, this is a gap in the present version of the argument.

Candidate resolution: Set $p = q = 1$, then we can remove the constraint $\alpha_k = 1/L_1$, and we still have the same results with some $\alpha_{\min} \in \{\alpha_{\mathcal{N}}^*, 1/L_1\}$.

7 Near-Saddle Analysis and Escape

The improve-or-localize lemma (Lemma 5) is generic and curvature-blind: by itself, it does not exclude the *localize* case at a strict saddle. To exclude it, we study the dynamics of $u_k := x_k - \tilde{x}$ *centered at the snapshot*, using the second-order Taylor expansion of f around $\tilde{x}$. The expansion is accurate in a small ball around $\tilde{x}$ — precisely the *localize* regime.

7.1 Step III — Linearized dynamics around the snapshot

Rationale. We linearize in order to perform a spectral analysis: get a linear system, compute its eigenvalues, and show that the system has an eigenvalue strictly larger than 1. The element of study is $u_k = x_k - \tilde{x}$, and specifically its projection $c_k = \langle u_k, v_{\min} \rangle$ on the negative-curvature direction: if the escape eigenvalue μ_+ exceeds 1, then c_k is amplified geometrically and the iterate is repelled from the stationary point along $v_{\min}$. Everything after this subsection exists to control the non-linear corrections to this picture.

Throughout this section, we condition on a perturbation trigger at iterate $\tilde{x}$, i.e.

$$\|g_{\tilde{x}}\| \leq \varepsilon_g, \quad \lambda_{\min}(\nabla^2 f(\tilde{x})) \leq -\varepsilon_H, \quad g_{\tilde{x}} := \nabla f(\tilde{x}).$$

We center the analysis at $\tilde{x}$. Set

$$H := \nabla^2 f(\tilde{x}),$$

and let $(\lambda_{\min}, v_{\min})$ denote the eigenpair of H at its most negative eigenvalue; by the trigger, $|\lambda_{\min}| \geq \varepsilon_H$. Define

$$u_k := x_k - \tilde{x}, \quad c_k := \langle u_k, v_{\min} \rangle, \quad \eta_k := \langle d_{k-1}, v_{\min} \rangle, \quad (15)$$

with the convention $d_{-1} := 0$ (so $\eta_0 = 0$, encoding the post-perturbation momentum reset $d_0 = -g_0$). Inside the window the stepsize is constant, $\alpha = 1/L_1$ (cf. Algorithm 1; but see Remark 5).

Quadratic model at $\tilde{x}$. The second-order Taylor expansion of f at $\tilde{x}$ is

$$\tilde{f}(x) = f(\tilde{x}) + g_{\tilde{x}}^\top (x - \tilde{x}) + \frac{1}{2} (x - \tilde{x})^\top H (x - \tilde{x}), \quad \nabla \tilde{f}(x) = g_{\tilde{x}} + H(x - \tilde{x}). \quad (16)$$

The full gradient differs from the model by the error

$$\tilde{e}(x) := \nabla f(x) - g_{\tilde{x}} - H(x - \tilde{x}), \quad (17)$$

bounded in Lemma 7 by $\|\tilde{e}(x)\| \leq (L_2/2) \|x - \tilde{x}\|^2$. We write $\tilde{e}_k := \tilde{e}(x_k)$. Unlike the expansion at a critical point, the constant term $g_{\tilde{x}}$ does *not* vanish: the snapshot has small ($\leq \varepsilon_g$) but nonzero gradient.

Linearized update. The NCG iteration $x_{k+1} = x_k + \alpha_k d_k$ with $d_k = -g_k + \beta_k d_{k-1}$ and in-window stepsizes $\alpha_k \in \{\alpha_{\min}, 1/L_1\}$ (Assumption 3), combined with the decomposition $g_k = g_{\tilde{x}} + H u_k + \tilde{e}_k$, gives

$$d_k = -H u_k + \beta_k d_{k-1} - g_{\tilde{x}} - \tilde{e}_k, \quad (18)$$

$$u_{k+1} = (I - \alpha_k H) u_k + \alpha_k \beta_k d_{k-1} - \alpha_k g_{\tilde{x}} - \alpha_k \tilde{e}_k. \quad (19)$$

Compared with a recurrence centered at an exact critical point, the new feature is the constant forcing $-\alpha_k g_{\tilde{x}}$.

State-space form. Stacking $z_k := (u_k, d_{k-1})$,

$$z_{k+1} = M_k z_k - b - r_k, \quad (20)$$

where

$$M_k = \begin{pmatrix} I - \alpha H & \alpha \beta_k I \\ -H & \beta_k I \end{pmatrix}, \quad b = \begin{pmatrix} \alpha g_{\tilde{x}} \\ g_{\tilde{x}} \end{pmatrix}, \quad r_k = \begin{pmatrix} \alpha \tilde{e}_k \\ \tilde{e}_k \end{pmatrix}. \quad (21)$$

The additive terms — the constant forcing b from $g_{\tilde{x}}$ and the model-error residual r_k — are exactly the corrections that the later steps will control.

Diagonalization. Since H is symmetric, it diagonalizes in an orthonormal eigenbasis $\{v_i\}_{i=1}^d$ with eigenvalues $\lambda_1 \leq \dots \leq \lambda_d$ (so $\lambda_1 = \lambda_{\min}$ and $v_1 = v_{\min}$). Projecting (20) onto each v_i decouples M_k into d independent 2×2 blocks

$$M_{k,i} = \begin{pmatrix} 1 - \alpha \lambda_i & \alpha \beta_k \\ -\lambda_i & \beta_k \end{pmatrix} \quad (22)$$

acting on the scalar pair $(\langle u_k, v_i \rangle, \langle d_{k-1}, v_i \rangle)$. The next lemma shows that, along the negative-curvature direction, $M_{k,i}$ has an eigenvalue strictly larger than 1.

Lemma 6 (Escape eigenvalue along negative curvature). *Fix a step k and an eigenpair (λ_i, v_i) of H with $\lambda_i < 0$, and suppose $\beta_k \geq 0$ (as holds for Fletcher-Reeves). The matrix $M_{k,i}$ in (22) has a real eigenvalue*

$$\mu_+ \geq 1 + \alpha |\lambda_i| > 1, \quad (23)$$

with equality $\mu_+ = 1 + \alpha |\lambda_i|$ iff $\beta_k = 0$ (the gradient-descent case).

Proof. The block $M_{k,i}$ has trace $M_{k,i} = (1 - \alpha \lambda_i) + \beta_k$ and

$$\det M_{k,i} = (1 - \alpha \lambda_i) \beta_k - (\alpha \beta_k)(-\lambda_i) = \beta_k.$$

Its characteristic polynomial is therefore $p(\mu) = \mu^2 - (1 - \alpha \lambda_i + \beta_k)\mu + \beta_k$, an upward parabola. Evaluating at $\mu = 1$,

$$p(1) = 1 - (1 - \alpha \lambda_i + \beta_k) + \beta_k = \alpha \lambda_i < 0,$$

since $\alpha > 0$ and $\lambda_i < 0$. An upward parabola negative at $\mu = 1$ has two distinct real roots straddling 1, so the larger root exceeds 1. For the sharper bound, set $w := \alpha|\lambda_i| = -\alpha\lambda_i > 0$, so that $p(\mu) = \mu^2 - (1 + w + \beta_k)\mu + \beta_k$ and

$$p(1 + w) = (1 + w)^2 - (1 + w + \beta_k)(1 + w) + \beta_k = -w\beta_k \leq 0.$$

Hence $1 + w$ lies between the two roots (or equals the larger one when $\beta_k = 0$), so the larger root satisfies $\mu_+ \geq 1 + w$, with equality iff $\beta_k = 0$. Explicitly, $\mu_{\pm} = \frac{1}{2}[(1 + w + \beta_k) \pm \sqrt{(1 + w + \beta_k)^2 - 4\beta_k}]$. $\square$

Projected recurrence on c_k . Projecting (19) on $v_{\min}$ and using $Hv_{\min} = \lambda_{\min}v_{\min}$,

$$c_{k+1} = (1 - \alpha_k \lambda_{\min}) c_k + \alpha_k \beta_k \eta_k - \alpha_k \langle g_{\tilde{x}}, v_{\min} \rangle - \alpha_k \langle \tilde{e}_k, v_{\min} \rangle. \quad (24)$$

Since $-\lambda_{\min} \geq \varepsilon_H$, we have $1 - \alpha_k \lambda_{\min} \geq 1 + \alpha_k \varepsilon_H$. The reverse triangle inequality applied to (24) yields

$$|c_{k+1}| \geq (1 + \alpha_k \varepsilon_H) |c_k| - \alpha_k \beta_k |\eta_k| - \alpha_k \varepsilon_g - \alpha_k \|\tilde{e}_k\|. \quad (25)$$

where we used $|\langle g_{\tilde{x}}, v_{\min} \rangle| \leq \|g_{\tilde{x}}\| \leq \varepsilon_g$ and $|\langle \tilde{e}_k, v_{\min} \rangle| \leq \|\tilde{e}_k\|$. The recurrence (25) contains three perturbing terms: the *momentum residual* $\alpha\beta_k|\eta_k|$, the *snapshot forcing* $\alpha\varepsilon_g$, and the *model error* $\alpha\|\tilde{e}_k\|$. Each will be controlled separately in the inductive analysis (Section 7.7).

Remark 6 (Interpretation, and what remains to be done). The escape eigenvalue $\mu_+ > 1$ from Lemma 6 drives geometric growth of c_k under the pure linearization (i.e., $\tilde{e} \equiv 0$ and $g_{\tilde{x}} = 0$): the iterate is repelled from the snapshot along the negative-curvature direction $v_{\min}$. In the actual NCG iteration, three corrections must be handled:

- the *model error* $\tilde{e}_k$, controlled by Lemma 7 (Step IV);
- the *snapshot forcing* $g_{\tilde{x}}$, small by the trigger condition $\|g_{\tilde{x}}\| \leq \varepsilon_g$;
- the *absence of an initial signal*: the geometric growth $|c_k| \gtrsim \mu_+^k |c_0|$ is vacuous if $|c_0|$ is zero (or tiny). The random perturbation ξ guarantees a nontrivial initial $|c_0| = |\langle \xi, v_{\min} \rangle|$ via the anti-concentration estimate of Lemma 8 (Step V).

The momentum residual additionally requires an *upper bound on β_k* , which for Fletcher–Reeves requires a gradient *lower* bound (Step VI). The remainder of Section 7 formalizes each of these corrections, and Step VII assembles them into the induction.

7.2 Step IV — Quadratic model error

Rationale. Straightforward step: bound $\|\tilde{e}_k\|$ above. This is the error between the real gradient and the quadratically approximated one; if it could exceed the geometric push $\alpha\varepsilon_H|c_k|$, the escape signal could be cancelled. The bound is quadratic in the distance to the snapshot, which is why the analysis must keep the iterates confined (invariant (i) of Step VII).

Lemma 7 (Quadratic model error at the snapshot). *Under Assumption 2, for every $x \in \mathbb{R}^d$,*

$$\nabla f(x) = g_{\tilde{x}} + H(x - \tilde{x}) + \tilde{e}(x), \quad \|\tilde{e}(x)\| \leq \frac{L_2}{2} \|x - \tilde{x}\|^2. \quad (26)$$

Proof. Let $x_t := \tilde{x} + t(x - \tilde{x})$ for $t \in [0, 1]$, so $\frac{dx_t}{dt} = x - \tilde{x}$. Set $\phi(t) := \nabla f(x_t)$; by the chain rule, $\phi'(t) = \nabla^2 f(x_t)(x - \tilde{x})$. By the fundamental theorem of calculus,

$$\nabla f(x) - g_{\tilde{x}} = \nabla f(x) - \nabla f(\tilde{x}) = \int_0^1 \phi'(t) dt = \int_0^1 \nabla^2 f(x_t)(x - \tilde{x}) dt.$$

Subtracting $\int_0^1 H(x - \tilde{x}) dt = H(x - \tilde{x})$,

$$\tilde{e}(x) = \int_0^1 (\nabla^2 f(x_t) - H)(x - \tilde{x}) dt.$$

Taking norms and using Assumption 2 together with $x_t - \tilde{x} = t(x - \tilde{x})$,

$$\|\tilde{e}(x)\| \leq \int_0^1 \|\nabla^2 f(x_t) - \nabla^2 f(\tilde{x})\| \|x - \tilde{x}\| dt \leq \int_0^1 L_2 t \|x - \tilde{x}\|^2 dt = \frac{L_2}{2} \|x - \tilde{x}\|^2. \quad \square$$

7.3 Setting for the probabilistic escape

The post-perturbation iterate is

$$x_0 := \tilde{x} + \xi, \quad \xi \sim \mathcal{U}(\mathcal{B}^d(0, R^*)),$$

so that $u_0 = \xi$ and $\|u_0\| \leq R^*$ deterministically. The escape window runs PR-NCG for $T = T_{\text{perturb}}$ steps with the algorithm's ordinary stepsize rule, so that $\alpha_k \in \{\alpha_{\min}, 1/L_1\}$ at every window step (Assumption 3); in particular $\alpha_k \geq \alpha_{\min}$ and $\alpha_k L_1 \leq 1$ throughout.

The three distinguished constants of the analysis are the dimensional factor, the *optimal perturbation radius*, and the *anti-concentration threshold*:

$$a := \sqrt{\frac{\pi}{2(d+2)}}, \quad R^* := \frac{a \varepsilon_H \delta}{L_2}, \quad s_\delta := a \delta R^* = \frac{a^2 \varepsilon_H \delta^2}{L_2} = \frac{\pi \varepsilon_H \delta^2}{2 L_2 (d+2)}. \quad (27)$$

The choice $R = R^*$ is justified by the optimization carried out in the proof of Lemma 9 (Step VI). Direct computation from (27) gives the exact identities

$$\frac{R^*}{s_\delta} = \frac{1}{a \delta}, \quad L_2 (R^*)^2 = \varepsilon_H s_\delta, \quad (28)$$

which will be used repeatedly in Step IX (the second identity is the *structural identity* of Lemma 12).

7.4 Step V — Anti-concentration of the perturbation

Rationale. Straightforward step: show that the perturbation has a non-negligible component along the negative-curvature direction, with high probability. This is the only probabilistic ingredient in the escape analysis; everything downstream is deterministic conditional on the good event $\mathcal{G}_0$.

Lemma 8 (Anti-concentration). *Let $\xi \sim \mathcal{U}(\mathcal{B}^d(0, R))$ and let $v \in \mathbb{R}^d$ be a unit vector. For every $s \in (0, R]$,*

$$\mathbb{P}(|\langle \xi, v \rangle| \leq s) \leq \frac{s}{R} \sqrt{\frac{2(d+2)}{\pi}}. \quad (29)$$

In particular, with $R = R^$ as in (27) and $s = s_\delta$, the event*

$$\mathcal{G}_0 := \{|\langle \xi, v_{\min} \rangle| \geq s_\delta\} \quad \text{satisfies} \quad \mathbb{P}(\mathcal{G}_0) \geq 1 - \delta.$$

Proof. Step 1: rotational invariance reduction. The Lebesgue measure on $\mathcal{B}^d(0, R)$ is invariant under any orthogonal $Q \in O(d)$, hence $Q\xi \stackrel{d}{=} \xi$. Choose Q with $Qe_1 = v$ (complete v to an orthonormal basis). Then

$$\langle \xi, v \rangle = \langle \xi, Qe_1 \rangle = \langle Q^\top \xi, e_1 \rangle, \quad Q^\top \xi \stackrel{d}{=} \xi,$$

so the distribution of $\langle \xi, v \rangle$ depends only on $\|v\| = 1$, not on the direction of v . Take $v = e_1$ and set $X := \xi_1$.

Step 2: marginal density of X . Write $\xi = (\xi_1, \xi_\perp)$ with $\xi_\perp \in \mathbb{R}^{d-1}$. By Fubini's theorem on the uniform density,

$$f_X(x) = \int_{\mathbb{R}^{d-1}} \frac{\mathbf{1}_{\{x^2 + \|\xi_\perp\|^2 \leq R^2\}}}{V_d R^d} d\xi_\perp,$$

with $V_k := \pi^{k/2}/\Gamma(k/2 + 1)$ the unit-ball volume in $\mathbb{R}^k$. For $|x| \leq R$, the slice is the $(d-1)$ -ball of radius $\sqrt{R^2 - x^2}$, of measure $V_{d-1}(R^2 - x^2)^{(d-1)/2}$, so

$$f_X(x) = \frac{V_{d-1}}{V_d R^d} (R^2 - x^2)^{(d-1)/2} \mathbf{1}_{\{|x| \leq R\}}. \quad (30)$$

Step 3: density peaks at 0. For $|x| \leq R$, $(R^2 - x^2)^{(d-1)/2}$ is decreasing in $|x|$, hence

$$f_X(x) \leq f_X(0) = \frac{V_{d-1}}{V_d R}.$$

Step 4: volume ratio via Gautschi's inequality. By definition,

$$\frac{V_{d-1}}{V_d} = \frac{1}{\sqrt{\pi}} \cdot \frac{\Gamma(d/2 + 1)}{\Gamma((d+1)/2)}.$$

Recall Gautschi's inequality: for $x > 0$ and $0 < s < 1$, $\Gamma(x+1)/\Gamma(x+s) \leq (x+1)^{1-s}$. With $x = d/2$, $s = 1/2$,

$$\frac{\Gamma(d/2 + 1)}{\Gamma((d+1)/2)} \leq \sqrt{(d+2)/2}, \quad \text{so} \quad \frac{V_{d-1}}{V_d} \leq \sqrt{\frac{d+2}{2\pi}}.$$

Step 5: integrate. Combining Steps 3 and 4, $f_X(0) \leq \frac{1}{R} \sqrt{(d+2)/(2\pi)}$, so

$$\mathbb{P}(|X| \leq s) = \int_{-s}^s f_X(x) dx \leq 2s f_X(0) \leq \frac{s}{R} \sqrt{\frac{2(d+2)}{\pi}},$$

which is (29).

Step 6: high-probability event. With $R = R^*$ and $s = s_\delta = a\delta R^*$, recalling $a = \sqrt{\pi/(2(d+2))}$,

$$\mathbb{P}(|\langle \xi, v_{\min} \rangle| \leq s_\delta) \leq \frac{s_\delta}{R^*} \sqrt{\frac{2(d+2)}{\pi}} = a\delta \cdot \frac{1}{a} = \delta,$$

so $\mathbb{P}(\mathcal{G}_0) = 1 - \mathbb{P}(|\langle \xi, v_{\min} \rangle| \leq s_\delta) \geq 1 - \delta$. □

Remark 7 (Reading Lemma 8). This lemma is the only probabilistic ingredient in the escape analysis. It guarantees that, with probability at least $1 - \delta$, the random perturbation has a non-negligible component along the negative-curvature direction: $|\langle \xi, v_{\min} \rangle| \geq s_\delta$. A priori the uniform perturbation could land almost orthogonally to $v_{\min}$, giving $\langle \xi, v_{\min} \rangle \approx 0$ and no usable escape signal, in which case the geometric growth from Lemma 6 would have nothing to amplify. Lemma 8 excludes this bad case with high probability; everything that follows is deterministic conditional on the event $\mathcal{G}_0$.

7.5 Step VI — Post-perturbation gradient lower bound

Rationale (Fletcher–Reeves specific). This step exists to control the momentum term. Since $\beta_k^{\text{FR}} = \|g_k\|^2 / \|g_{k-1}\|^2$, bounding β^{FR} requires bounding $\|g_k\|$ *above* and $\|g_{k-1}\|$ *below*: $\beta^{\text{FR}} \leq \bar{g}^2 / \underline{g}^2$. The upper bound is easy (Lipschitz, cf. (38)); the lower bound is the hard part, because Lipschitz continuity only ever produces upper bounds, and the window sits in a

region where the gradient can in principle vanish. Any other β formula would mean redoing this step; heavy ball or Nesterov do not need it, because their momentum coefficient is constant. This subsection establishes the lower bound at time 0 (and, as a by-product, fixes the optimal perturbation radius R^*); propagating the lower bound across the whole window is the job of the induction (Step VII).

Lemma 9 (Post-perturbation gradient lower bound). *Suppose*

$$\varepsilon_g < \frac{1}{2} \varepsilon_H s_\delta. \quad (31)$$

Then on the event $\mathcal{G}_0$ of Lemma 8,

$$\|g_0\| \geq \underline{g}_0, \quad \underline{g}_0 := \frac{1}{2} \varepsilon_H s_\delta - \varepsilon_g > 0.$$

Proof. Step 1: quadratic-model decomposition. By Lemma 7 applied at $x_0 = \tilde{x} + \xi$ and $u_0 = \xi$,

$$g_0 = g_{\tilde{x}} + H \xi + \tilde{e}_0, \quad \|\tilde{e}_0\| \leq \frac{L_2}{2} \|\xi\|^2 \leq \frac{L_2}{2} R^2,$$

where R is the perturbation radius (to be optimized in Step 5 below).

Step 2: project on $v_{\min}$. Using $H v_{\min} = \lambda_{\min} v_{\min}$,

$$\langle g_0, v_{\min} \rangle = \underbrace{\langle g_{\tilde{x}}, v_{\min} \rangle}_{=:C} + \underbrace{\lambda_{\min} \langle \xi, v_{\min} \rangle}_{=:P} + \underbrace{\langle \tilde{e}_0, v_{\min} \rangle}_{=:E}. \quad (32)$$

Step 3: bound the three terms. On $\mathcal{G}_0$, $|\langle \xi, v_{\min} \rangle| \geq s_\delta$; combined with $|\lambda_{\min}| \geq \varepsilon_H$,

$$|P| \geq \varepsilon_H s_\delta, \quad |C| \leq \|g_{\tilde{x}}\| \leq \varepsilon_g, \quad |E| \leq \|\tilde{e}_0\| \leq \frac{L_2}{2} R^2.$$

Step 4: combine via reverse triangle inequality. From $\langle g_0, v_{\min} \rangle = C + P + E$ and $|a + b + c| \geq |a| - |b| - |c|$,

$$\|g_0\| \geq |\langle g_0, v_{\min} \rangle| \geq |P| - |C| - |E| \geq \varepsilon_H s_\delta - \varepsilon_g - \frac{L_2}{2} R^2. \quad (33)$$

Note $s_\delta = a\delta R$ depends on R ; substituting,

$$\|g_0\| \geq h(R) - \varepsilon_g, \quad h(R) := a\varepsilon_H \delta R - \frac{L_2}{2} R^2. \quad (34)$$

The map $R \mapsto h(R)$ is a downward parabola.

Step 5: optimize in R . Setting $h'(R) = 0$,

$$h'(R) = a\varepsilon_H \delta - L_2 R \implies R^* = \frac{a\varepsilon_H \delta}{L_2},$$

matching (27). Compute $h(R^*)$ in two pieces:

$$(R^*)^2 = \frac{a^2 \varepsilon_H^2 \delta^2}{L_2^2}, \quad \frac{L_2}{2} (R^*)^2 = \frac{a^2 \varepsilon_H^2 \delta^2}{2 L_2} = \frac{1}{2} \varepsilon_H s_\delta, \quad a\varepsilon_H \delta R^* = \frac{a^2 \varepsilon_H^2 \delta^2}{L_2} = \varepsilon_H s_\delta,$$

where we used $s_\delta = a^2 \varepsilon_H \delta^2 / L_2$ from (27). (Note in passing that the middle computation is the structural identity $L_2 (R^*)^2 = \varepsilon_H s_\delta$ of (49).) Summing,

$$h(R^*) = \varepsilon_H s_\delta - \frac{1}{2} \varepsilon_H s_\delta = \frac{1}{2} \varepsilon_H s_\delta.$$

Step 6: conclude. Setting $R = R^*$ in (34),

$$\|g_0\| \geq \frac{1}{2} \varepsilon_H s_\delta - \varepsilon_g = \underline{g}_0,$$

strictly positive under the smallness condition (31). $\square$

Remark 8 (Why this lemma matters). If $\|g_0\|$ could vanish, the algorithm could trigger another perturbation immediately and the whole escape window would be wasted; more importantly, the FR coefficient β_1 would be uncontrolled. Lemma 9 rules this out at time 0. The window-wide floor $\underline{g}$ of (39), established inductively in Lemma 11, is the time- k analogue; note that $\underline{g} \leq \underline{g}_0$, since the window floor pays the model error at the larger radius $S + R^*$ rather than R^* (compare (39) with $\underline{g}_0 = \varepsilon_H s_\delta - \varepsilon_g - \frac{L_2}{2} (R^*)^2$, using the structural identity).

7.6 Step VII — Setting up the escape window

Rationale. The single perturbation has placed the iterate at $x_0 = \tilde{x} + \xi$, and the algorithm now runs for T steps. Our goal is to prove that, with high probability, the function value drops by at least F at some point within this window. Following the improve-or-localize dichotomy (Remark 4), there are two cases:

- (a) the function value drops by at least F somewhere in $\{0, \dots, T\}$ — this is what we try to prove;
- (b) the function value never drops by that much, in which case Lemma 5 tells us the iterates stay close to x_0 ($\|x_k - x_0\| \leq S$, $S = \sqrt{2FT/L_1}$).

We work in case (b) and derive a contradiction, so we must be in case (a). Case (b) is the bad case, and we formalize it by the localization event below.

The localization event. Let $T \in \mathbb{N}$ and $F > 0$ be parameters (their values will be fixed in Theorem 1 and Step IX). Define the *localization event*

$$\mathcal{L}_T := \left\{ \max_{0 \leq k \leq T} (f(x_0) - f(x_k)) < F \right\}. \quad (35)$$

The event $\mathcal{L}_T$ is the bad case: the function value has not dropped by F anywhere in the window. Its complement is exactly the conclusion we want. So the strategy is to show that $\mathcal{G}_0 \cap \mathcal{L}_T$ is empty: throughout this section and the next, we work conditional on $\mathcal{G}_0 \cap \mathcal{L}_T$ and derive a contradiction; combined with $\mathbb{P}(\mathcal{G}_0) \geq 1 - \delta$ this gives the result.

Confinement on $\mathcal{L}_T$. On $\mathcal{L}_T$ the improve-or-localize lemma (Lemma 5) forces the iterates to stay near x_0 . Recentering at the snapshot yields the confinement we need.

Lemma 10 (Localized confinement around $\tilde{x}$). *Set the localization radius*

$$S := \sqrt{\frac{2FT}{L_1}}. \quad (36)$$

On the event $\mathcal{L}_T$, the displacement from the snapshot satisfies

$$\|u_k\| = \|x_k - \tilde{x}\| \leq S + R^* \quad \text{for every } k \in \{0, 1, \dots, T\}. \quad (37)$$

Proof. Fix $k \in \{0, \dots, T\}$; the case $k = 0$ is immediate from $\|u_0\| = \|\xi\| \leq R^*$. Writing the displacement as a telescoping sum, then applying the triangle inequality and Cauchy–Schwarz on the sum of k terms,

$$\|x_k - x_0\|^2 = \left\| \sum_{j=0}^{k-1} (x_{j+1} - x_j) \right\|^2 \leq \left(\sum_{j=0}^{k-1} \|x_{j+1} - x_j\| \right)^2 \leq k \sum_{j=0}^{k-1} \|x_{j+1} - x_j\|^2.$$

Every window step is either restarted or a non-restarted step with α_N^* (Assumption 3, Remark 2), so the per-step bound (13) of Lemma 5 applies at every j ; its right-hand sides are nonnegative and telescope:

$$\sum_{j=0}^{k-1} \|x_{j+1} - x_j\|^2 \leq \frac{2}{L_1} (f(x_0) - f(x_k)) < \frac{2F}{L_1} \quad \text{on } \mathcal{L}_T.$$

Combining, and using $k \leq T$, $\|x_k - x_0\|^2 < 2kF/L_1 \leq 2TF/L_1 = S^2$. The triangle inequality with $\|x_0 - \tilde{x}\| = \|\xi\| \leq R^*$ gives $\|u_k\| \leq \|x_k - x_0\| + \|x_0 - \tilde{x}\| \leq S + R^*$. $\square$

The two contributions to $S + R^*$ are independent: S tracks how far the iterate may wander inside the localization ball, while R^* accounts for the initial perturbation displacement. This is why the analysis is posed with respect to $S + R^*$ rather than S alone. (Caveat: the application of Lemma 5 to in-window steps is subject to the gap flagged in Remark 5.)

Validity of the quadratic model. At distance ρ from $\tilde{x}$, the negative-curvature signal of $\nabla^2 f(\tilde{x})$ contributes roughly $\varepsilon_H \rho$ to the gradient in direction $v_{\min}$, while the quadratic-model error of Lemma 7 grows as $\frac{L_2}{2} \rho^2$. For the signal to dominate the error throughout the confinement ball, we need $\varepsilon_H \rho \geq \frac{L_2}{2} \rho^2$ at the worst point $\rho = S + R^*$. This is the validity condition

$$S + R^* \leq \frac{2\varepsilon_H}{L_2}. \quad (\text{C-valid})$$

Beyond this threshold, the Lipschitz-Hessian variation swamps the curvature signal and the linear model is no longer reliable. It will turn out (??) that (C-valid) is *implied* by the constraint (C-err) of Lemma 11, so it does not need to be carried separately in the constants ledger of Step IX.

Uniform gradient upper bound. The momentum analysis requires bounding the FR coefficient $\beta_k = \|g_k\|^2 / \|g_{k-1}\|^2$, which in turn requires both an upper and a lower bound on gradient norms across the window. The upper bound is immediate from gradient-Lipschitz: applying Assumption 1 to the pair $(x_k, \tilde{x})$ and combining with the trigger bound $\|\nabla f(\tilde{x})\| \leq \varepsilon_g$ and the confinement (37),

$$\|g_k\| \leq \|g_{\tilde{x}}\| + L_1 \|x_k - \tilde{x}\| \leq \varepsilon_g + L_1 (S + R^*) =: \bar{g} \quad \text{for every } k \in \{0, \dots, T\} \text{ on } \mathcal{L}_T. \quad (38)$$

The first term ε_g accounts for the residual gradient at the snapshot; the second term $L_1(S + R^*)$ accounts for the additional gradient that develops as the iterate moves through the confinement ball. In the relevant regime $\varepsilon_g \ll L_1(S + R^*)$, the second term dominates.

The complementary lower bound cannot come from Lipschitz alone — Lipschitz gives only upper bounds. In fact *no* bound over the confinement ball can supply it: the ball is centered at a near-critical point, so $\inf_{x \in \mathcal{B}(\tilde{x}, S+R^*)} \|\nabla f(x)\|$ is essentially zero. The lower bound is a property of *where the iterates actually go*, not of the region they are in; it requires that $|c_k|$ stays large and that the model error stays small, both of which are properties we establish in the induction. We therefore defer the lower bound to that section, where it appears as the gradient-floor invariant.

Gradient-floor placeholder. Anticipating the induction, define the gradient floor

$$\underline{g} := \varepsilon_H s_\delta - \varepsilon_g - \frac{L_2}{2} (S + R^*)^2. \quad (39)$$

We will show in Lemma 11 that $\underline{g} > 0$ under the budget-split constraints, and that $\|g_k\| \geq \underline{g}$ for every k in the window on $\mathcal{G}_0 \cap \mathcal{L}_T$. Until then, $\underline{g}$ is a notational placeholder for the quantity we will eventually establish as a uniform lower bound. Together, $\bar{g}$ and $\underline{g}$ yield the FR envelope

$$\beta_k = \frac{\|g_k\|^2}{\|g_{k-1}\|^2} \leq \frac{\bar{g}^2}{\underline{g}^2} =: \bar{\beta}, \quad (40)$$

used in the momentum constraint (C-mom) below.

Budget splits for the geometric driver. The recurrence (25) for c_k has the shape $|c_{k+1}| \geq (1 + \alpha\varepsilon_H)|c_k| -$ (three perturbing terms), with contributions from the snapshot forcing $g_{\tilde{x}}$, the quadratic-model error $\tilde{e}_k$, and the momentum term $\beta_k \eta_k$. To extract geometric growth, we will require each of these to be a controlled fraction of the driving term $\alpha\varepsilon_H |c_k|$. Introduce three positive splits $\theta, \theta', \theta'' \in (0, 1)$ with $\theta + \theta' + \theta'' < 1$, and define the residual growth rate

$$\gamma := 1 - \theta - \theta' - \theta''. \quad (41)$$

The values of the splits are *not* fixed here: they are decision variables of the analysis. Their optimal values are derived exactly in the complexity section (Definition 3); until then every statement holds for any admissible choice.

Summary of the setup. We have committed to: a localization event $\mathcal{L}_T$ on which we work; a localization radius S and a confinement bound $S + R^*$ around $\tilde{x}$; a quadratic-model validity constraint (C-valid) (later absorbed into (C-err)); an upper bound $\bar{g}$ on gradient norms; budget splits $\theta, \theta', \theta''$; and a placeholder gradient floor $\underline{g}$. The next subsection states the joint invariants and proves them by induction.

7.7 Step VII — Joint invariants on the escape window

Rationale: why an induction is unavoidable. The recurrence (25),

$$|c_{k+1}| \geq (1 + \alpha\varepsilon_H) |c_k| - \alpha\varepsilon_g - \alpha\beta_k |\eta_k| - \alpha \|\tilde{e}_k\|,$$

cannot be iterated directly, because the obstacle terms depend on the past:

- $\|\tilde{e}_k\| \leq \frac{L_2}{2} \|u_k\|^2$ depends on $u_k = x_k - \tilde{x}$;
- β_k depends on $\|g_{k-1}\|$, which depends on $\|u_{k-1}\|$ and on $|c_{k-1}|$;
- $|\eta_k| = |\langle d_{k-1}, v_{\min} \rangle|$ depends on the previous directions.

The solution is an induction, which lets us maintain a set of properties simultaneously up to time k , and prove them at time $k + 1$. From the recurrence, the three things we need are:

1. how far the iterate is: $\|x_k - \tilde{x}\| \leq S + R^*$;
2. how big the gradient is: $\|g_k\| \geq \underline{g}$;
3. how big $|c_k|$ is: $|c_k| \geq (1 + \gamma\alpha\varepsilon_H)^k |c_0|$.

It is worth pausing on why no shortcut exists for item 2. The upper envelope $\bar{g}$ is a supremum over the confinement ball — a static quantity, valid for any point of the ball. The lower bound has no static analogue: the infimum of $\|\nabla f\|$ over a ball centered at a near-critical point is essentially zero. The floor holds only along the trajectory, and only *because* the trajectory carries a large escape component: $\|g_k\| \geq \varepsilon_H |c_k| - \varepsilon_g - \|\tilde{e}_k\|$. This creates the circular dependency

$$\text{floor at } j < k \implies \beta_j \text{ controlled} \implies \text{growth of } |c_j| \text{ up to } k \implies \text{floor at } k,$$

which is broken precisely by carrying all three properties together in the inductive hypothesis: each property at step $k + 1$ is purchased with a *different* property at step k . Without (1) the model error blows up; without (2) the FR coefficient β_k blows up; without (3) there is no floor and no escape. They prop each other up.

The three constraints (C-snap), (C-err), (C-mom) below each have the form “*this perturbing source is bounded by $\theta_\bullet \cdot \varepsilon_H s_\delta$* ”. Combined with the inductive lower bound $|c_k| \geq s_\delta$, each constraint absorbs the corresponding perturbing term into a $\theta_\bullet$ -fraction of the budget $\alpha\varepsilon_H |c_k|$, leaving the residual fraction $\gamma = 1 - \theta - \theta' - \theta''$ as the geometric growth rate.

Lemma 11 (Window invariants for the escape). *Let $p = q = 1$ (Assumption 3) and suppose the three budget-split constraints*

$$\varepsilon_g \leq \theta'' \varepsilon_H s_\delta \quad (\text{C-snap}) \quad (42)$$

$$\frac{L_2}{2} (S + R^*)^2 \leq \theta \varepsilon_H s_\delta \quad (\text{C-err}) \quad (43)$$

$$\bar{\beta} \kappa \bar{g} \leq \theta' \varepsilon_H s_\delta \quad (\text{C-mom}) \quad (44)$$

are satisfied. Then $\underline{g} > 0$ and, on $\mathcal{G}_0 \cap \mathcal{L}_T$, for every $k \in \{0, \dots, T\}$:

- (i) $\|u_k\| \leq S + R^*$ (confinement),

(ii) $\|g_k\| \geq \underline{g}$ (gradient floor),

(iii) $|c_k| \geq \prod_{j=0}^{k-1} (1 + \gamma \alpha_j \varepsilon_H) |c_0| \geq (1 + \gamma \alpha_{\min} \varepsilon_H)^k |c_0|$ with $|c_0| \geq s_\delta$ (geometric growth).

Proof. Positivity of $\underline{g}$. By (39) and the constraints (C-snap), (C-err),

$$\underline{g} = \varepsilon_H s_\delta - \varepsilon_g - \frac{L_2}{2} (S + R^*)^2 \geq (1 - \theta - \theta'') \varepsilon_H s_\delta > 0,$$

since $\theta + \theta'' < 1$. The remainder of the proof is by induction on k : the inductive hypothesis at step k is that (i)–(iii) hold at every index $j \in \{0, \dots, k\}$ (strong form; this is what makes β_k , which involves $\|g_{k-1}\|$, controllable).

Base case ($k = 0$). *Confinement.* $u_0 = \xi$, so $\|u_0\| = \|\xi\| \leq R^* \leq S + R^*$.

Growth. $c_0 = \langle \xi, v_{\min} \rangle$ and the event $\mathcal{G}_0$ is defined precisely as $|c_0| \geq s_\delta$.

Floor. Project the model identity (26) at x_0 on $v_{\min}$ and apply the reverse triangle inequality with $|\lambda_{\min}| \geq \varepsilon_H$, the snapshot trigger $\|g_{\tilde{x}}\| \leq \varepsilon_g$, and the growth bound just established:

$$\|g_0\| \geq |\lambda_{\min}| |c_0| - \|g_{\tilde{x}}\| - \|\tilde{e}_0\| \geq \varepsilon_H s_\delta - \varepsilon_g - \frac{L_2}{2} (R^*)^2 \geq \underline{g},$$

where the last step uses $(R^*)^2 \leq (S + R^*)^2$ in the definition of $\underline{g}$.

Inductive step. Assume invariants (i), (ii), (iii) hold at every index $j \leq k$ for some $0 \leq k < T$. We establish them at step $k + 1$ in the order: (i) confinement, then growth (iii), then floor (ii). Each sub-claim uses specific tools, which we identify line-by-line.

Confinement at $k + 1$. *Claim.* $\|u_{k+1}\| \leq S + R^*$.

Why this is direct from Lemma 10, not from the recurrence. Lemma 10 establishes $\|u_j\| \leq S + R^*$ uniformly for every $j \in \{0, \dots, T\}$ on $\mathcal{L}_T$. The proof of that lemma uses improve-or-localize applied to the entire window — a global descent identity. Since we are conditioning on $\mathcal{L}_T$ and $k + 1 \leq T$, the bound (37) applies at index $k + 1$ with no further work:

$$\|u_{k+1}\| \leq S + R^* \quad \text{by (37) at } j = k + 1.$$

Note that this step does *not* use any inductive hypothesis at k . Confinement is not propagated step-by-step. Remark 9 explains why: the projection $|c_{k+1}| \leq \|u_{k+1}\|$ only recovers the $v_{\min}$ -component, so step-local control of $|c_{k+1}|$ alone cannot bound $\|u_{k+1}\|$.

Setting up the recurrence. The recurrence (25), evaluated at index k , reads

$$|c_{k+1}| \geq (1 + \alpha_k \varepsilon_H) |c_k| - \alpha_k \varepsilon_g - \alpha_k \beta_k |\eta_k| - \alpha_k \|\tilde{e}_k\|. \quad (45)$$

We bound each of the three negative terms by a fraction $\theta_\bullet$ of the geometric driver $\alpha_k \varepsilon_H |c_k|$. The key observation is that the constraints (C-snap), (C-err), (C-mom) are α_k -free: each absorption is proved *before* multiplying by the stepsize, and is then multiplied by the *same* α_k on both sides, so variable stepsizes cost nothing here.

Bound on snapshot forcing. *Claim.* $\alpha_k \varepsilon_g \leq \theta'' \alpha_k \varepsilon_H |c_k|$.

Tools. (C-snap), inductive hypothesis (iii) at k .

Argument.

$$\begin{aligned} \alpha_k \varepsilon_g &\leq \alpha_k \theta'' \varepsilon_H s_\delta && \text{(multiply (C-snap) by } \alpha_k) \\ &\leq \alpha_k \theta'' \varepsilon_H |c_k| && \text{(by (iii): } |c_k| \geq |c_0| \geq s_\delta). \end{aligned}$$

Bound on quadratic-model error. *Claim.* $\alpha_k \|\tilde{e}_k\| \leq \theta \alpha_k \varepsilon_H |c_k|$.

Tools. Lemma 7 (eq. (26)), inductive hypothesis (i) at k , (C-err), inductive hypothesis (iii) at k .

Argument. By Lemma 7 applied at x_k , $\|\tilde{e}_k\| \leq \frac{L_2}{2} \|u_k\|^2$. The IH (i) at k gives $\|u_k\| \leq S + R^*$, so

$$\|\tilde{e}_k\| \leq \frac{L_2}{2} (S + R^*)^2.$$

Now (C-err) bounds the right-hand side by $\theta \varepsilon_H s_\delta$, and (iii) at k upgrades s_δ to $|c_k|$:

$$\|\tilde{e}_k\| \leq \frac{L_2}{2} (S + R^*)^2 \leq \theta \varepsilon_H s_\delta \leq \theta \varepsilon_H |c_k|.$$

Multiplying by α_k :

$$\alpha_k \|\tilde{e}_k\| \leq \theta \alpha_k \varepsilon_H |c_k|.$$

Bound on the momentum term. *Claim.* $\alpha_k \beta_k |\eta_k| \leq \theta' \alpha_k \varepsilon_H |c_k|$.

Tools. The algorithm's restart test (Algorithm 1, with $q = 1$), gradient upper bound (38), gradient floor (ii) at indices $k - 1$ and k , (C-mom), IH (iii) at k .

Argument. First bound $|\eta_k|$:

$$|\eta_k| = |\langle d_{k-1}, v_{\min} \rangle| \leq \|d_{k-1}\|.$$

By the algorithm's restart test (with $q = 1$, $\kappa \geq 1$), $\|d_{k-1}\| \leq \kappa \|g_{k-1}\|$ regardless of whether restart fired:

- If restart fired at $k - 1$: $d_{k-1} = -g_{k-1}$, so $\|d_{k-1}\| = \|g_{k-1}\| \leq \kappa \|g_{k-1}\|$ (using $\kappa \geq 1$).
- If restart did not fire: the contrapositive of the restart condition gives $\|d_{k-1}\| < \kappa \|g_{k-1}\|$.

Combined with $\|g_{k-1}\| \leq \bar{g}$ from (38):

$$|\eta_k| \leq \kappa \|g_{k-1}\| \leq \kappa \bar{g}.$$

For β_k , the FR formula together with the upper bound (38) at index k and the gradient floor at index $k - 1$ (which we have by the strong IH (ii)) gives $\beta_k = \|g_k\|^2 / \|g_{k-1}\|^2 \leq \bar{g}^2 / \underline{g}^2 = \bar{\beta}$. Then

$$\beta_k |\eta_k| \leq \bar{\beta} \kappa \bar{g}.$$

By (C-mom), this is at most $\theta' \varepsilon_H s_\delta$; by (iii) at k , $s_\delta \leq |c_k|$, so

$$\beta_k |\eta_k| \leq \bar{\beta} \kappa \bar{g} \leq \theta' \varepsilon_H s_\delta \leq \theta' \varepsilon_H |c_k|.$$

Multiplying by α_k :

$$\alpha_k \beta_k |\eta_k| \leq \theta' \alpha_k \varepsilon_H |c_k|.$$

Edge case at $k = 0$. The convention $d_{-1} = 0$ encoding the post-perturbation momentum reset gives $\eta_0 = \langle d_{-1}, v_{\min} \rangle = 0$, so the bound holds trivially at $k = 0$ without invoking (C-mom) (and without a floor at index -1).

Growth at $k + 1$. Substituting the three bounds into (45):

$$\begin{aligned} |c_{k+1}| &\geq (1 + \alpha_k \varepsilon_H) |c_k| - \theta'' \alpha_k \varepsilon_H |c_k| - \theta' \alpha_k \varepsilon_H |c_k| - \theta \alpha_k \varepsilon_H |c_k| \\ &= (1 + \gamma \alpha_k \varepsilon_H) |c_k|, \end{aligned}$$

using $\gamma = 1 - \theta - \theta' - \theta''$. Combined with IH (iii) at k and $\alpha_k \geq \alpha_{\min}$:

$$|c_{k+1}| \geq \prod_{j=0}^k (1 + \gamma \alpha_j \varepsilon_H) |c_0| \geq (1 + \gamma \alpha_{\min} \varepsilon_H)^{k+1} |c_0|,$$

with $|c_0| \geq s_\delta$ on $\mathcal{G}_0$ (preserved from the base case).

Floor at $k + 1$. *Claim.* $\|g_{k+1}\| \geq \underline{g}$.

Tools. Lemma 7 at x_{k+1} , trigger bound on $\|g_{\tilde{x}}\|$, confinement (i) at $k + 1$ (just established), growth (iii) at $k + 1$ (just established), definition of $\underline{g}$ (39).

Argument. Lemma 7 gives $\nabla f(x_{k+1}) = g_{\tilde{x}} + H u_{k+1} + \tilde{e}_{k+1}$ where $H = \nabla^2 f(\tilde{x})$. Projecting on $v_{\min}$:

$$\langle g_{k+1}, v_{\min} \rangle = \langle g_{\tilde{x}}, v_{\min} \rangle + \lambda_{\min} c_{k+1} + \langle \tilde{e}_{k+1}, v_{\min} \rangle.$$

Since $\|g_{k+1}\| \geq |\langle g_{k+1}, v_{\min} \rangle|$ (Cauchy–Schwarz, $v_{\min}$ unit vector), the reverse triangle inequality gives

$$\|g_{k+1}\| \geq |\lambda_{\min}| \cdot |c_{k+1}| - \|g_{\tilde{x}}\| - \|\tilde{e}_{k+1}\|.$$

Now plug in:

- $|\lambda_{\min}| \geq \varepsilon_H$ (trigger condition);
- $|c_{k+1}| \geq s_\delta$ (growth (iii) at $k + 1$, combined with $|c_0| \geq s_\delta$);
- $\|g_{\tilde{x}}\| \leq \varepsilon_g$ (trigger condition);
- $\|\tilde{e}_{k+1}\| \leq \frac{L_2}{2} \|u_{k+1}\|^2 \leq \frac{L_2}{2} (S + R^*)^2$ (Lemma 7 and confinement (i) at $k + 1$).

This yields

$$\|g_{k+1}\| \geq \varepsilon_H s_\delta - \varepsilon_g - \frac{L_2}{2} (S + R^*)^2 = \underline{g},$$

by the definition (39), which completes the induction. $\square$

Remark 9 (Why step-local confinement fails). The projection $|c_{k+1}| \leq \|u_{k+1}\|$ recovers only the $v_{\min}$ -component of the displacement; bounding that component does not control the orthogonal components. Confinement at step $k + 1$ therefore cannot be deduced from the recurrence (25) alone — it requires the global descent identity (Lemma 5), which enters through the conditioning on $\mathcal{L}_T$ in (35).

7.8 Step VIII — Escape theorem

Rationale. Using invariant (iii) of the induction we show that $|c_T|$ exceeds $(1 + \gamma \alpha_k \varepsilon_H)^T s_\delta$; using invariant (i) we know $|c_T| \leq \|u_T\| \leq S + R^*$. Choosing T so that the first quantity exceeds the second produces a contradiction, hence $\mathcal{G}_0 \cap \mathcal{L}_T = \emptyset$: on the good perturbation event, the localization case is impossible, so the function value must drop by at least F within the window.

Theorem 1 (Single-trajectory saddle escape). *Define*

$$\Phi := \log\left(\frac{S + R^*}{s_\delta}\right) \quad (46)$$

and choose the window length

$$T := \left\lceil \frac{\Phi}{\log(1 + \gamma \alpha_{\min} \varepsilon_H)} \right\rceil + 1. \quad (47)$$

Under the assumptions of Lemma 11,

$$\mathbb{P}\left(f(x_0) - \min_{0 \leq k \leq T} f(x_k) \geq F\right) \geq \mathbb{P}(\mathcal{G}_0) \geq 1 - \delta.$$

Proof. We show $\mathcal{G}_0 \cap \mathcal{L}_T = \emptyset$. Suppose, for contradiction, that a sample point lies in $\mathcal{G}_0 \cap \mathcal{L}_T$. By Lemma 11(iii),

$$|c_T| \geq (1 + \gamma \alpha_{\min} \varepsilon_H)^T s_\delta.$$

Since $\lceil y \rceil + 1 > y$ for every y , the choice (47) gives

$$T \log(1 + \gamma \alpha_{\min} \varepsilon_H) > \Phi, \quad \text{hence} \quad (1 + \gamma \alpha_{\min} \varepsilon_H)^T s_\delta > e^\Phi s_\delta = S + R^*,$$

so $|c_T| > S + R^*$. On the other hand, by Lemma 11(i) and Cauchy–Schwarz, $|c_T| \leq \|u_T\| \leq S + R^*$. Contradiction. Therefore on $\mathcal{G}_0$ the event $\mathcal{L}_T$ fails, i.e. the function value drops by at least F within the escape window; since $\mathbb{P}(\mathcal{G}_0) \geq 1 - \delta$ by Lemma 8, the conclusion follows. $\square$

8 Setting and the constraint system

Throughout, L_1 is the gradient Lipschitz constant, L_2 the Hessian Lipschitz constant, $\varepsilon_g, \varepsilon_H > 0$ the tolerances, d the dimension, κ the momentum constant from the restart condition, $\delta \in (0, 1)$ the per-window failure probability, and $\mathcal{L} := f(x_0^{\text{alg}}) - f_{\text{low}}$ the total decrease budget. Define

$$a := \sqrt{\frac{\pi}{2(d+2)}}, \quad R^* := \frac{a \varepsilon_H \delta}{L_2}, \quad s_\delta := \frac{a^2 \varepsilon_H \delta^2}{L_2}, \quad (48)$$

the perturbation radius and the guaranteed initial projection on the escape direction (probability $\geq 1 - \delta$). Direct computation from (48) gives the exact identities

$$\frac{R^*}{s_\delta} = \frac{1}{a \delta}, \quad L_2 (R^*)^2 = \varepsilon_H s_\delta. \quad (49)$$

The escape-window analysis (Sections 7.2–7.5 of the main document) is valid provided the parameters S (localization radius from the post-perturbation start), T (window length), F (per-window decrease threshold) and the splits $\theta, \theta', \theta'', \gamma > 0$ satisfy the following system, in which $\bar{g} := \varepsilon_g + L_1(S + R^*)$ and $\underline{g} := \varepsilon_H s_\delta - \varepsilon_g - \frac{L_2}{2}(S + R^*)^2$ are the upper and lower gradient envelopes over the window, α_k is the in-window stepsize satisfying $\alpha_k \in [\alpha_{\min}, 1/L_1]$ with $\alpha_{\min} = \sigma/(L_1 \kappa^2)$, and $\bar{\beta}$ the momentum-coefficient envelope ($\bar{\beta} \leq \bar{g}^2/\underline{g}^2$ for Fletcher–Reeves):

$$(\text{C-snap}) : \quad \varepsilon_g \leq \theta'' \varepsilon_H s_\delta, \quad (50)$$

$$(\text{C-err}) : \quad \frac{L_2}{2} (S + R^*)^2 \leq \theta \varepsilon_H s_\delta, \quad (51)$$

$$(\text{C-mom}) : \quad \bar{\beta} \kappa \bar{g} \leq \theta' \varepsilon_H s_\delta, \quad (52)$$

$$(\text{C-win}) : \quad T \geq \frac{\log((S + R^*)/s_\delta)}{\log(1 + \gamma \alpha_{\min} \varepsilon_H)}, \quad (53)$$

together with the budget constraint

$$\theta + \theta' + \theta'' + \gamma \leq 1. \quad (54)$$

Define the fixed quantity

$$\theta''_{\min} := \frac{\varepsilon_g}{\varepsilon_H s_\delta} \quad (\text{so } (\text{C-snap}) \text{ reads } \theta'' \geq \theta''_{\min}), \quad (55)$$

and the dimensionless *regime parameter*

$$\lambda := \frac{L_1 R^*}{\varepsilon_H s_\delta} \stackrel{(49)}{=} \frac{L_1}{a \delta \varepsilon_H} = \frac{L_1}{\delta \varepsilon_H} \sqrt{\frac{2(d+2)}{\pi}}. \quad (56)$$

Small λ means the perturbation geometry is large relative to the gradient-Lipschitz scale; “deep regime” below means $\lambda \rightarrow 0$.

Finally, normalize the localization radius (we write ς , reserving σ for the algorithm’s sufficient-descent constant, which enters Step IX through $\alpha_{\min} = \sigma/(L_1 \kappa^2)$):

$$\varsigma := \frac{L_1 S}{\varepsilon_H s_\delta}, \quad \text{so that} \quad \frac{L_1 (S + R^*)}{\varepsilon_H s_\delta} = \varsigma + \lambda. \quad (57)$$

9 The two envelope bounds on S

Lemma 12 (Structural identity and lower bound on θ). *For every $S \geq 0$, (C-err) implies $\theta \geq 1/2$, with strict inequality whenever $S > 0$.*

Proof. By (49), $\frac{L_2}{2}(R^*)^2 = \frac{1}{2}\varepsilon_H s_\delta$ exactly. Since $S \geq 0$, $\frac{L_2}{2}(S + R^*)^2 \geq \frac{L_2}{2}(R^*)^2 = \frac{1}{2}\varepsilon_H s_\delta$, so (51) forces $\theta \varepsilon_H s_\delta \geq \frac{1}{2}\varepsilon_H s_\delta$, i.e. $\theta \geq 1/2$. If $S > 0$ the first inequality is strict. $\square$

Lemma 13 (Error envelope (B-err)). (C-err) is equivalent to

$$S + R^* \leq \sqrt{2\theta} R^*, \quad \text{i.e.} \quad \varsigma \leq \sigma_{\text{err}}(\theta) := (\sqrt{2\theta} - 1) \lambda. \quad (58)$$

Proof. By (49), $\frac{L_2}{2}(S + R^*)^2 \leq \theta \varepsilon_H s_\delta \iff (S + R^*)^2 \leq 2\theta(R^*)^2 \iff S + R^* \leq \sqrt{2\theta} R^*$ (both sides nonnegative). Multiplying by $L_1/(\varepsilon_H s_\delta)$ and using (56), (57) gives $\varsigma + \lambda \leq \sqrt{2\theta} \lambda$. $\square$

Lemma 14 (Momentum envelope (B-mom), Fletcher–Reeves). *Suppose (C-err) holds and $B := 1 - \theta - \theta''_{\min} > 0$. Then a sufficient condition for (C-mom) with $\bar{\beta} \leq \bar{g}^2/\underline{g}^2$ is*

$$\varsigma \leq \sigma_{\text{mom}}(\theta, \theta') := \left(\frac{\theta' B^2}{\kappa} \right)^{1/3} - \theta''_{\min} - \lambda. \quad (59)$$

Proof. Using $\varepsilon_g = \theta''_{\min} \varepsilon_H s_\delta$ (definition (55)) and (51),

$$\underline{g} = \varepsilon_H s_\delta - \varepsilon_g - \frac{L_2}{2}(S + R^*)^2 \geq \varepsilon_H s_\delta (1 - \theta''_{\min} - \theta) = B \varepsilon_H s_\delta. \quad (60)$$

Since $x \mapsto \kappa \bar{g}^3/x^2$ is decreasing on $x > 0$, the lower bound (60) gives the *upper* bound $\bar{\beta} \kappa \bar{g} \leq \kappa \bar{g}^3/\underline{g}^2 \leq \kappa \bar{g}^3/(B \varepsilon_H s_\delta)^2$ — note the direction: bounding $\underline{g}$ from *below* is what makes the substitution sufficient rather than merely necessary. It therefore suffices for (52) that

$$\frac{\kappa \bar{g}^3}{B^2(\varepsilon_H s_\delta)^2} \leq \theta' \varepsilon_H s_\delta \iff \bar{g} \leq \left(\frac{\theta' B^2}{\kappa} \right)^{1/3} \varepsilon_H s_\delta.$$

With $\bar{g} = \varepsilon_g + L_1(S + R^*) = (\theta''_{\min} + \varsigma + \lambda) \varepsilon_H s_\delta$ by (55), (57), this is exactly (59). $\square$

The admissible localization radius is therefore

$$\boxed{\varsigma \leq \min \{ \sigma_{\text{err}}(\theta), \sigma_{\text{mom}}(\theta, \theta') \}.} \quad (61)$$

Lemma 15 (Necessary regime condition). *If $\varsigma > 0$ is admissible for some splits satisfying (54), then*

$$\lambda + \theta''_{\min} < \frac{B}{\kappa^{1/3}} \leq \frac{1}{2\kappa^{1/3}}. \quad (62)$$

Proof. $\sigma_{\text{mom}} > 0$ requires $(\theta' B^2/\kappa)^{1/3} > \theta''_{\min} + \lambda$. Since $\theta' \leq B$ by (54) ($\theta' + \gamma \leq B$ and $\gamma > 0$), $(\theta' B^2/\kappa)^{1/3} \leq B/\kappa^{1/3}$, and $B = 1 - \theta - \theta''_{\min} \leq 1/2$ by Lemma 12. $\square$

Unpacking (56), condition (62) requires $\varepsilon_H \delta > 2\kappa^{1/3} L_1 \sqrt{2(d+2)/\pi}$: the regime restriction of the single-trajectory analysis. It is *unchanged* by the corrected optimization below; only the value of S inside the regime changes.

10 Optimization of the splits

10.1 The objective and its exact surrogate

The quantities derived in Section 11 give $N_{\text{iter}} \leq 2T^2\mathcal{L}/(L_1S^2)+T$ with $T \leq \Phi/(\gamma \alpha_{\min} \varepsilon_H)+\Phi+2$ and $\Phi = \log((S+R^*)/s_\delta)$. We therefore *maximize*

$$J := \gamma \varsigma \quad (63)$$

over the splits, subject to (61), (54), $\theta \geq 1/2$, $\theta'' \geq \theta''_{\min}$.

Remark 10 (Exactness of the surrogate). N_{iter} and J are related by $N_{\text{iter}} \lesssim T^2/S^2 \propto 1/(\gamma\varsigma)^2 = 1/J^2$ up to three explicitly bounded discrepancies: (i) $\log(1+x) \in [x/(1+x), x]$, so under $\gamma \alpha_{\min} \varepsilon_H \leq 1$ the bound on T is exact within a factor 2, hence N_{iter} within a factor 4; (ii) Φ depends on the splits only through $\log(S+R^*)$, a logarithmic variation; (iii) the additive terms $+2, +T$. Consequently, maximizing J minimizes the N_{iter} bound up to an explicit absolute factor; combined with Proposition 1 this yields $N_{\text{iter}}^{\text{canonical}} \leq 16 N_{\text{iter}}^{\text{opt}}$ under (R'). No claim sharper than this factor is made.

10.2 Structure of the optimum

Lemma 16 (θ'' is pinned). *At any maximizer of J with $J > 0$, $\theta'' = \theta''_{\min}$.*

Proof. θ'' enters J only through the budget (54): neither σ_{err} nor σ_{mom} depends on θ'' (the quantity $\theta''_{\min}$ in (59) is the fixed problem datum (55), not the split θ'' , and B is defined with $\theta''_{\min}$). If $\theta'' > \theta''_{\min}$, replace (θ'', γ) by $(\theta''_{\min}, \gamma + \theta'' - \theta''_{\min})$: all constraints still hold and J strictly increases. $\square$

In view of Lemma 16 we set $\theta'' = \theta''_{\min}$, write

$$M := 1 - \theta''_{\min}, \quad B = M - \theta, \quad \gamma = B - \theta', \quad (64)$$

and saturate the budget (54) (any slack is given to γ , which strictly increases J). The remaining variables are $(\theta, \theta') \in [1/2, M] \times (0, B)$.

Lemma 17 (Balance at the optimum). *Assume $J^* := \sup J > 0$. Then the supremum is attained, and at every maximizer*

$$\sigma_{\text{err}}(\theta) = \sigma_{\text{mom}}(\theta, \theta'), \quad \text{i.e.} \quad \left(\frac{\theta' B^2}{\kappa}\right)^{1/3} = \theta''_{\min} + \sqrt{2\theta} \lambda. \quad (65)$$

Proof. J is continuous on the compact set $\{\theta \in [1/2, M], \theta' \in [0, M-\theta]\}$ (with $\varsigma := \min\{\sigma_{\text{err}}, \sigma_{\text{mom}}\}^+$), so a maximizer exists.

Case $\sigma_{\text{mom}} > \sigma_{\text{err}}$ at a maximizer. Then $\sigma_{\text{mom}} > \sigma_{\text{err}} \geq 0$ forces $\theta' > 0$. Decrease θ' by $\eta > 0$ and give η to γ . By continuity of σ_{mom} in θ' , for η small enough still $\sigma_{\text{mom}} \geq \sigma_{\text{err}}$, so $\varsigma = \sigma_{\text{err}}$ is unchanged while γ strictly increases: J strictly increases, contradiction.

Case $\sigma_{\text{mom}} < \sigma_{\text{err}}$ at a maximizer. Then $J > 0$ forces $\sigma_{\text{mom}} > 0$, and $\sigma_{\text{err}} > \sigma_{\text{mom}} > 0$ forces $\sqrt{2\theta} > 1$, i.e. $\theta > 1/2$ strictly. Decrease θ by $\eta > 0$ and give η to γ (keeping θ'): B increases, so σ_{mom} strictly increases; σ_{err} decreases continuously, so for η small enough still $\sigma_{\text{err}} > \sigma_{\text{mom}}$. Hence $\varsigma = \sigma_{\text{mom}}$ strictly increased and γ strictly increased: J strictly increases, contradiction.

Therefore $\sigma_{\text{err}} = \sigma_{\text{mom}}$ at every maximizer; the displayed equivalent form follows by adding $\theta''_{\min} + \lambda$ to $\sigma_{\text{err}} = (\sqrt{2\theta} - 1)\lambda$ and cubing. $\square$

10.3 Exact upper bound and the canonical splits

Lemma 18 (Exact upper bound on the objective). *Let $M > 1/2$ and*

$$t_\star := \frac{1 + \sqrt{1 + 6M}}{3} \in (1, \sqrt{2M}). \quad (66)$$

Then for every feasible choice of splits,

$$J \leq \lambda t_\star (t_\star - 1)^2. \quad (67)$$

Proof. For any feasible point, $J = \gamma\sigma \leq \gamma\sigma_{\text{err}} \leq B\sigma_{\text{err}} = (M - \theta)(\sqrt{2\theta} - 1)\lambda$, using $\gamma = B - \theta' \leq B$. Substitute $t := \sqrt{2\theta} \in [1, \sqrt{2M}]$ and define

$$h(t) := \left(M - \frac{t^2}{2}\right)(t - 1), \quad h'(t) = -\frac{3}{2}t^2 + t + M, \quad h''(t) = -3t + 1 < 0 \quad (t \geq 1).$$

h is strictly concave on $[1, \sqrt{2M}]$ with $h'(1) = M - \frac{1}{2} > 0$ and $h'(\sqrt{2M}) = \sqrt{2M} - 2M < 0$ (since $M > 1/2$), so h has a unique interior maximizer, the positive root of h' , which is (66). At $t_\star$ the first-order condition gives the exact identity

$$M = \frac{3t_\star^2 - 2t_\star}{2} \implies M - \frac{t_\star^2}{2} = t_\star^2 - t_\star = t_\star(t_\star - 1), \quad (68)$$

hence $h(t_\star) = t_\star(t_\star - 1)^2$, which is (67). $\square$

Definition 3 (Canonical splits). With $t_\star$ as in (66), set

$$\theta_\star := \frac{t_\star^2}{2}, \quad \theta_\star'' := \theta_{\min}'', \quad B_\star := M - \theta_\star \stackrel{(68)}{=} t_\star(t_\star - 1), \quad \theta_\star' := \frac{\kappa(\theta_{\min}'' + t_\star\lambda)^3}{B_\star^2}, \quad \gamma_\star := B_\star - \theta_\star', \quad (69)$$

and take σ at its admissible maximum, $\sigma_\star := (t_\star - 1)\lambda$, i.e.

$$S = (t_\star - 1)R^* = (t_\star - 1)\sqrt{\frac{\pi}{2(d+2)}} \frac{\varepsilon_H \delta}{L_2}, \quad S + R^* = t_\star R^*. \quad (70)$$

Lemma 19 (Feasibility of the canonical splits). *Assume $\theta_{\min}'' < 1/2$ (tolerance compatibility, equivalently $\varepsilon_g < \frac{1}{2}\varepsilon_H s_\delta$) and the regime restriction*

$$\theta_{\min}'' + t_\star\lambda \leq \frac{B_\star}{(2\kappa)^{1/3}}. \quad (\text{R}') \quad (71)$$

Then the canonical splits are feasible: (C-snap), (C-err), (C-mom) all hold with S as in (70), $\theta_\star' \leq B_\star/2$, and $\gamma_\star \geq B_\star/2 > 0$. Moreover (C-err) and (C-mom) are saturated (hold with equality in the envelope bounds (58), (59)).

Proof. $\theta_{\min}'' < 1/2$ gives $M > 1/2$, so $t_\star \in (1, \sqrt{2M})$ and $\theta_\star = t_\star^2/2 \in (1/2, M)$, $B_\star > 0$. (R') cubed reads $\kappa(\theta_{\min}'' + t_\star\lambda)^3 \leq B_\star^3/2$, i.e. $\theta_\star' \leq B_\star/2$, whence $\gamma_\star \geq B_\star/2 > 0$ and (54) holds with equality by construction. (C-err): by (70) and (49), $\frac{L_2}{2}(S + R^*)^2 = \frac{t_\star^2}{2}\varepsilon_H s_\delta = \theta_\star \varepsilon_H s_\delta$ — equality. (C-mom): $\bar{g} = (\theta_{\min}'' + t_\star\lambda)\varepsilon_H s_\delta$ by (57), and by (60), $\bar{g} \geq B_\star \varepsilon_H s_\delta$, so

$$\bar{\beta} \kappa \bar{g} \leq \frac{\kappa \bar{g}^3}{\bar{g}^2} \leq \frac{\kappa(\theta_{\min}'' + t_\star\lambda)^3}{B_\star^2} \varepsilon_H s_\delta = \theta_\star' \varepsilon_H s_\delta,$$

with equality in the last step by definition of $\theta_\star'$: (52) holds and the envelope (59) is saturated, i.e. $\sigma_{\text{mom}}(\theta_\star, \theta_\star') = \sigma_\star = \sigma_{\text{err}}(\theta_\star)$, which is the balance equation (65). $\square$

Proposition 1 (Near-optimality, exact form). *Let*

$$\rho_\star := \frac{\theta'_\star}{B_\star} = \kappa \left(\frac{\theta''_{\min} + t_\star \lambda}{t_\star(t_\star - 1)} \right)^3. \quad (71)$$

Under the assumptions of Lemma 19,

$$J_{\text{can}} = \gamma_\star \sigma_\star = (1 - \rho_\star) \lambda t_\star (t_\star - 1)^2 \geq (1 - \rho_\star) \sup J, \quad \rho_\star \leq \frac{1}{2}. \quad (72)$$

In particular $J_{\text{can}} \geq \frac{1}{2} \sup J$ always under (R'), and $\rho_\star \rightarrow 0$ as $\lambda, \theta''_{\min} \rightarrow 0$: deep in the regime the canonical splits are asymptotically exactly optimal.

Proof. $J_{\text{can}} = (B_\star - \theta'_\star)(t_\star - 1)\lambda = (1 - \rho_\star)B_\star(t_\star - 1)\lambda = (1 - \rho_\star)t_\star(t_\star - 1)^2\lambda$ by (68). Combine with the upper bound (67), valid for every feasible point. The bound $\rho_\star \leq 1/2$ is (R') cubed. $\square$

Remark 11 (Why the previous resolution is invalid). The earlier derivation fixed $\theta = 1/2$ (“with infinitesimal margin”), declared (C-mom) the binding constraint, and evaluated S from saturated (B-mom), obtaining $S + R^* = B \varepsilon_H s_\delta / (L_1(4\kappa)^{1/3})$ and hence $S \propto \delta^2$. By Lemma 13, however, $\sigma_{\text{err}}(1/2) = 0$: at $\theta = 1/2$, (C-err) forces $S = 0$, and near $\theta = 1/2$, $S \leq (\sqrt{2\theta} - 1)R^* \approx (\theta - \frac{1}{2})R^*$ scales *linearly* in the margin. The value of S used in the previous Steps 6–10 violates (C-err) whenever it is positive, deep in the regime $\lambda \ll 1$ — which is precisely where it was invoked. The binding structure is in fact: (C-mom) binds only near the boundary of (R'); deep in the regime both constraints bind simultaneously at the optimum (Lemma 17), with $S = (t_\star - 1)R^* \propto \delta$, not δ^2 . Consequently the previously stated per-window count $N_{\text{iter}} \propto L_1^3 L_2^2 d^2 \mathcal{L} / (\varepsilon_H^6 \delta^4)$ does not correspond to any feasible instantiation of the constraint system and must be retracted; the feasible count is Theorem 2 below, which exceeds it by the factor λ^{-2} (up to absolute constants) inside the regime, the two coinciding at the boundary of (R').

Remark 12 (Anchor point and the radius $S + 2R^*$). All four constraints above are anchored at the *snap point*, where the trigger certifies $\lambda_{\min}(\nabla^2 f(\text{snap})) \leq -\varepsilon_H$; the confinement radius from the snap point is $S + R^*$ (perturbation $\leq R^*$, in-window travel $\leq S$), and this is what enters (C-err) and $\bar{g}$. If instead (C-err) is stated with radius $S + \nu R^*$ for $\nu \geq 1$ (e.g. $\nu = 2$ when anchoring at the underlying saddle, at distance up to R^* from the snap point), the identity of Lemma 12 becomes $\frac{L_2^2}{2}(\nu R^*)^2 / (\varepsilon_H s_\delta) = \nu^2/2$, forcing $\theta \geq \nu^2/2$; for $\nu = 2$ this gives $\theta \geq 2$, incompatible with (54): the system is infeasible. The infeasibility cannot be repaired by shrinking the perturbation radius alone: if the perturbation radius is R^*/ϱ ($\varrho \geq 1$) while the snap-to-saddle distance remains $\leq R^*$, then s_δ scales by $1/\varrho$ and the saddle-anchored identity becomes $\varrho(1+1/\varrho)^2/2 = (\varrho+2+1/\varrho)/2 \geq 2$ for all $\varrho \geq 1$. Hence saddle-anchoring of (C-err) is structurally infeasible under the present definitions (48); the analysis must remain snap-anchored, which is also what the trigger certifies.

11 Derived quantities: S, T, F, N_{iter}

Throughout this section the canonical splits (69) are in force, S is given by (70), and the in-window stepsizes satisfy $\alpha_k \in \{\sigma/(L_1\kappa^2), 1/L_1\} \subset [\alpha_{\min}, 1/L_1]$ (Assumption 3), with $\alpha_{\min} = \sigma/(L_1\kappa^2)$.

Window length. Set

$$\Phi := \log\left(\frac{S + R^*}{s_\delta}\right) \stackrel{(70),(49)}{=} \log\left(\frac{t_\star}{a\delta}\right) = \log\left(\frac{t_\star}{\delta} \sqrt{\frac{2(d+2)}{\pi}}\right), \quad T := \left\lceil \frac{\Phi}{\log(1 + \gamma_\star \alpha_{\min} \varepsilon_H)} \right\rceil + 1, \quad (73)$$

so that (C-win) holds by construction; this is the choice (47) of Theorem 1 with $\gamma = \gamma_\star$ and $S + R^* = t_\star R^*$. Using $\log(1+x) \geq x/(1+x)$ for $x \geq 0$,

$$T \leq \frac{\Phi}{\log(1 + \gamma_\star \alpha_{\min} \varepsilon_H)} + 2 \leq \frac{(1 + \gamma_\star \alpha_{\min} \varepsilon_H) \Phi}{\gamma_\star \alpha_{\min} \varepsilon_H} + 2 = \frac{L_1 \kappa^2 \Phi}{\sigma \gamma_\star \varepsilon_H} + \Phi + 2. \quad (\text{T})$$

Per-window decrease. By Lemma 10, the localization radius on the no-decrease event is $S = \sqrt{2FT/L_1}$: the constant $2/L_1$ is the *uniform* improve-or-localize constant of Lemma 5, valid for every in-window step under Assumption 3 (restarted steps and non-restarted steps with $\alpha_{\mathcal{N}}^*$ are the only two iteration types). Equivalently, choosing

$$F := \frac{L_1 S^2}{2T} = \frac{(t_\star - 1)^2 \pi L_1 \varepsilon_H^2 \delta^2}{4(d+2) T L_2^2} \quad (\text{F})$$

makes the localization radius exactly S .

Iteration count (window iterations). Lifetime accounting: the number of perturbation windows satisfies $N_{\text{trig}} \leq \mathcal{L}/F + 1$, hence

$$N_{\text{iter}} \leq T N_{\text{trig}} \leq \frac{2T^2 \mathcal{L}}{L_1 S^2} + T = \frac{4(d+2) T^2 L_2^2 \mathcal{L}}{\pi L_1 (t_\star - 1)^2 \varepsilon_H^2 \delta^2} + T. \quad (\text{N})$$

Remark 13 (Role of the in-window stepsizes). By (T), $T \propto 1/\alpha_{\min}$ to leading order, so (N) scales as $N_{\text{iter}} \propto T^2 \propto \alpha_{\min}^{-2} = (L_1 \kappa^2 / \sigma)^2$: the price of taking genuine NCG steps inside the window, rather than gradient steps at $1/L_1$, is the absolute factor κ^4 / σ^2 . Since (σ, κ) are algorithm constants, this does not affect the dependence on $(\varepsilon_H, d, \delta)$. Outside the perturbation window the stepsize remains a free placeholder, unconstrained by this analysis.

Corollary 1 (Explicit count). *Assume additionally $\gamma_\star \varepsilon_H \leq L_1$ (which implies $\gamma_\star \alpha_{\min} \varepsilon_H \leq \sigma / \kappa^2 \leq 1$) and $a \delta \leq t_\star / e$ (so $\Phi \geq 1$). Then*

$$T \leq \frac{4 L_1 \kappa^2 \Phi}{\sigma \gamma_\star \varepsilon_H}$$

and

$$N_{\text{iter}} \leq \frac{64(d+2) \kappa^4 L_1 L_2^2 \Phi^2 \mathcal{L}}{\pi \sigma^2 \gamma_\star^2 (t_\star - 1)^2 \varepsilon_H^4 \delta^2} + \frac{4 L_1 \kappa^2 \Phi}{\sigma \gamma_\star \varepsilon_H}. \quad (74)$$

Proof. By (T), $T \leq L_1 \kappa^2 \Phi / (\sigma \gamma_\star \varepsilon_H) + \Phi + 2$. Since $\Phi \geq 1$, $\Phi + 2 \leq 3\Phi$; since $\gamma_\star \alpha_{\min} \varepsilon_H \leq 1$, $\Phi \leq \Phi / (\gamma_\star \alpha_{\min} \varepsilon_H) = L_1 \kappa^2 \Phi / (\sigma \gamma_\star \varepsilon_H)$. Hence $T \leq 4 L_1 \kappa^2 \Phi / (\sigma \gamma_\star \varepsilon_H)$. Substituting $T^2 \leq 16 L_1^2 \kappa^4 \Phi^2 / (\sigma^2 \gamma_\star^2 \varepsilon_H^2)$ and $S^2 = (t_\star - 1)^2 \pi \varepsilon_H^2 \delta^2 / (2(d+2) L_2^2)$ into (N) gives (74). $\square$

12 Union bound over windows

Each window fails (the perturbation misses the escape direction) with probability at most δ . On the event that all of the first $\lceil \mathcal{L}/F \rceil$ windows succeed, the algorithm performs at most $N_{\text{trig}} \leq \mathcal{L}/F + 1$ windows; by the union bound the failure probability is at most $(\mathcal{L}/F + 1) \delta$. Writing $F = c' \delta^2$ with

$$c' := \frac{\pi (t_\star - 1)^2 L_1 \varepsilon_H^2}{4(d+2) T L_2^2} \quad (\text{which depends on } \delta \text{ only through } \Phi(\delta) \text{ inside } T), \quad (75)$$

the requirement of total failure probability at most $\Delta \in (0, 1)$ reads

$$\frac{\mathcal{L}}{c'(\delta) \delta} + \delta \leq \Delta. \quad (\text{U})$$

Lemma 20 (Admissible interval for δ). *A sufficient condition for (U) is*

$$\max \left\{ \delta_{\text{reg}}, \delta_{\text{u}} \right\} \leq \delta \leq \frac{\Delta}{2}, \quad \text{where} \quad \delta_{\text{u}} \text{ solves } \delta_{\text{u}} = \frac{2 \mathcal{L}}{c'(\delta_{\text{u}}) \Delta}, \quad (76)$$

and δ_{reg} is the smallest δ satisfying the regime restriction (R'), namely

$$\delta_{\text{reg}} = \frac{t_\star (2\kappa)^{1/3} L_1}{a \varepsilon_H (B_\star - (2\kappa)^{1/3} \theta''_{\min})}, \quad \text{requiring} \quad B_\star > (2\kappa)^{1/3} \theta''_{\min}. \quad (77)$$

The threshold δ_u exists and is unique: the map $\delta \mapsto 2\mathcal{L}/(c'(\delta)\Delta)$ is strictly decreasing (as δ increases, Φ decreases, T decreases, c' increases), so it crosses the identity exactly once on $(0, 1)$. The interval (76) is nonempty if and only if $\delta_{\text{reg}} \leq \Delta/2$ and $\delta_u \leq \Delta/2$; the latter is equivalent to $c'(\Delta/2) \Delta^2 \geq 4\mathcal{L}$.

Proof. If $\delta \leq \Delta/2$ and $\delta \geq 2\mathcal{L}/(c'(\delta)\Delta)$, then $\mathcal{L}/(c'\delta) \leq \Delta/2$ and (U) follows by summing. Monotonicity of c' in δ : $\Phi(\delta) = \log(t_\star/(a\delta))$ is strictly decreasing; T in (73) is nondecreasing in Φ ; $c' \propto 1/T$. The crossing claim follows since the identity map is strictly increasing from 0 and the decreasing map is positive. Nonemptiness: $\delta \geq \delta_u$ for $\delta \leq \Delta/2$ is equivalent to $\Delta/2 \geq \delta_u$ by monotonicity, which unfolds to $c'(\Delta/2) (\Delta/2) \geq 2\mathcal{L}/\Delta$. $\square$

Remark 14 (Choice of δ within the admissible interval). The bound (N) is strictly decreasing in δ (as $1/\delta^2$ up to the logarithm in T), so within (76) the iteration bound is minimized at the upper endpoint $\delta = \Delta/2$.

Remark 15 (Structural change versus the previous coupling equation). Because $F \propto \delta^2$ (not δ^4), the self-consistency relation for δ is the degree-one crossing of Lemma 20, not the previous cubic $\delta^3 = 2T(4\kappa)^{2/3} L_1 L_2^2 \mathcal{L}/(B^2 a^4 \varepsilon_H^4 \Delta)$. That equation inherited the exponent 4 from the infeasible $S \propto \delta^2$ and is superseded.

13 Main theorem

Theorem 2 (PR–NCG escape: corrected exact constants). *Fix tolerances $\varepsilon_g, \varepsilon_H > 0$ and a global failure probability $\Delta \in (0, 1)$. Define a , $R^\star$, s_δ , $\theta''_{\min}$, λ by (48), (55), (56); $M = 1 - \theta''_{\min}$ and $t_\star$ by (66); the canonical splits by (69); and let δ be any value in the admissible interval (76). Assume:*

1. **Tolerance compatibility:** $\theta''_{\min} < 1/2$ and $B_\star > (2\kappa)^{1/3} \theta''_{\min}$;
2. **Regime restriction** (R'): $\theta''_{\min} + t_\star \lambda \leq B_\star / (2\kappa)^{1/3}$ (equivalently $\delta \geq \delta_{\text{reg}}$ of (77));
3. **Union-bound compatibility:** the interval (76) is nonempty;
4. the escape-window analysis (Lemma 11, Theorem 1) holds under (50)–(53), with $p = q = 1$ (Assumption 3) and in-window stepsizes $\alpha_k \in \{\sigma/(L_1 \kappa^2), 1/L_1\}$.

Set $S = (t_\star - 1)R^\star$, T as in (73), and $F = L_1 S^2 / (2T)$. Then with probability at least $1 - \Delta$, every perturbation window either decreases f by at least F or escapes the ε_H -saddle, the total number of windows is at most $\mathcal{L}/F + 1$, and the total number of window iterations satisfies the exact bound (N); under the additional assumptions of Corollary 1 ($\gamma_\star \varepsilon_H \leq L_1$, $a\delta \leq t_\star/e$),

$$N_{\text{iter}} \leq \frac{64(d+2)\kappa^4 L_1 L_2^2 \Phi^2 \mathcal{L}}{\pi \sigma^2 \gamma_\star^2 (t_\star - 1)^2 \varepsilon_H^4 \delta^2} + \frac{4 L_1 \kappa^2 \Phi}{\sigma \gamma_\star \varepsilon_H}, \quad \Phi = \log\left(\frac{t_\star}{\delta} \sqrt{\frac{2(d+2)}{\pi}}\right).$$

Moreover the canonical splits are optimal for the surrogate objective $J = \gamma\varsigma$ within the exact factor $(1 - \rho_\star) \geq 1/2$ of Proposition 1, hence optimal for the iteration bound within an explicit absolute factor (Remark 10); and by Lemma 15 no choice of splits removes the regime restriction $\lambda + \theta''_{\min} < 1/(2\kappa^{1/3})$, i.e. $\varepsilon_H \delta > 2\kappa^{1/3} L_1 \sqrt{2(d+2)/\pi}$.

Proof. Feasibility of the splits and saturation of (C-err), (C-mom) is Lemma 19; (C-win) holds by the definition of T in (73); (C-snap) holds by (55) and $\theta''_{\star} = \theta''_{\min}$. Assumption 4 then yields the per-window dichotomy with failure probability at most δ per window (Theorem 1). The window count and (N) are the lifetime accounting of Section 11; the probability statement is Lemma 20. Near-optimality is Proposition 1 with Remark 10, and the irremovability of the regime restriction is Lemma 15, which used only (54) and Lemma 12, i.e. holds for *every* feasible choice of splits. $\square$

Remark 16 (Scaling summary and comparison). Per window (fixed δ), the corrected count scales as $N_{\text{iter}} = \tilde{O}(\kappa^4 \sigma^{-2} L_1 L_2^2 (d+2) \mathcal{L} \varepsilon_H^{-4} \delta^{-2})$, valid only under (R'); the factor κ^4/σ^2 is an absolute constant of the algorithm (Remark 13). Against Jin–Netrapalli–Jordan (ε_H^{-4} , polylog d , no regime restriction), the remaining gaps are the $(d+2) \delta^{-2}$ factor and the regime restriction itself; both originate in the single-trajectory anti-concentration constant $a \propto d^{-1/2}$ and the Fletcher–Reeves envelope $\bar{g}/g$ (the cubic in Lemma 14), respectively, and neither is touched by any choice of splits. Removing them is the object of the coupling extension.

References

- [1] Rémi Chan–Renous–Legoubin and Clément W. Royer. A nonlinear conjugate gradient method with complexity guarantees and its application to nonconvex regression. *EURO Journal on Computational Optimization*, 10:100044, 2022.
- [2] Chi Jin, Rong Ge, Praneeth Netrapalli, Sham M. Kakade, and Michael I. Jordan. How to escape saddle points efficiently. In Doina Precup and Yee Whye Teh, editors, *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pages 1724–1732. PMLR, 06–11 Aug 2017.
- [3] Chi Jin, Praneeth Netrapalli, and Michael I. Jordan. Accelerated gradient descent escapes saddle points faster than gradient descent. In Sébastien Bubeck, Vianney Perchet, and Philippe Rigollet, editors, *Proceedings of the 31st Conference On Learning Theory*, volume 75 of *Proceedings of Machine Learning Research*, pages 1042–1085. PMLR, 06–09 Jul 2018.
- [4] Jason D. Lee, Ioannis Panageas, Georgios Piliouras, Max Simchowitz, Michael I. Jordan, and Benjamin Recht. First-order methods almost always avoid strict saddle points. *Mathematical Programming*, 176(1–2):311–337, 2019.